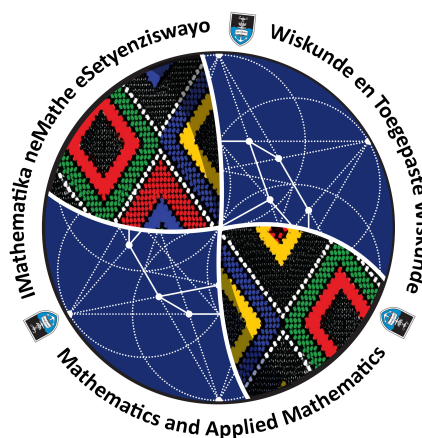


3DM: Discrete Mathematics



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A note on sources:

The exercises in these notes, and many of the proofs, have largely been taken from existing sources, including Michael Henning's notes [Hen05] and books by Gary Chartrand [Cha85], Gary Chartrand and Linda Lesniak [CL96], and Frank Harary [Har69].

A note on conventions:

- Most graph theorists don't use the 'real' real numbers (like π , $\sqrt{3}$, and $121/55$) very often. We do frequently work with integers, however. Throughout these notes, we therefore use the notation $[a, b]$ to indicate an *integer interval* rather than a closed interval, e.g., by $[3, 7]$ we mean the set $\{3, 4, 5, 6, 7\}$ rather than the set $\{x \in \mathbb{R} : 3 \leq x \leq 7\}$. Similarly, we shall use the notation $(3, 7)$ for the set $\{4, 5, 6\}$.
- To indicate the end of the proof of a *claim*, which is usually only part of the proof of a more substantial result, we use the symbol ©. For an example, see the proof of the Havel-Hakimi Theorem (Theorem 3).

Part I

Graph Theory

Chapter 1

Introduction

1.1 Graphs

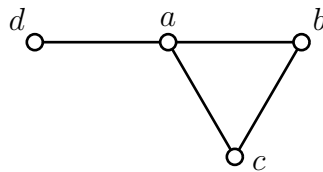
A *graph* G consists of:

1. A finite, nonempty set $V(G)$ of *vertices*¹, and,
2. A (possibly empty) set $E(G)$ of unordered pairs of vertices called *edges*.

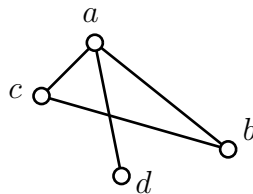
The set $V(G)$ is the *vertex set* of G . The set $E(G)$ is the *edge set* of G . Each edge is a 2-subset of the vertex set and so should be denoted with set notation. For convenience, however, we denote the edge joining the vertices a and b as ab rather than $\{a, b\}$. Since edges are unordered pairs of vertices, the edge ab is the same as the edge ba .

Graphs are frequently represented by pictures.

Example 1.1 Let G be the graph with $V(G) = \{a, b, c, d\}$ and $E(G) = \{ab, ac, ad, bc\}$. Here is one way to draw G :



Here is another picture of the same graph G :



Both of these pictures represent the same graph.

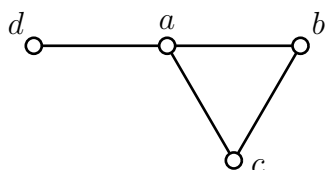
Some more notation: In the graph G , the edge ab *joins* the vertices a and b . Since a and b are joined by an edge, they are *adjacent* vertices, and a is said to be a *neighbour* of b . The *neighbourhood* of b is the set of all neighbours of b , i.e., the neighbourhood of b is the set $\{a, c\}$.

¹The singular of vertices is *vertex*.

The vertices c and d are *nonadjacent*. The edge ab is *incident* with the vertices a and b , while the vertex a is *incident* with the edges ab , ac , and ad .

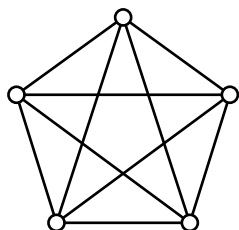
The *order* of a graph is the number of vertices, while the *size* of a graph is the number of edges. Usually, we use n for order and m for size. A graph of order n and size m is an (n, m) graph. If we want to make it clear that we mean the order and size of G (rather than some other graph), then we write $n(G)$ and $m(G)$.

Example 1.2 The graph G from Example 1.1 has order $n = 4$ and size $m = 4$:



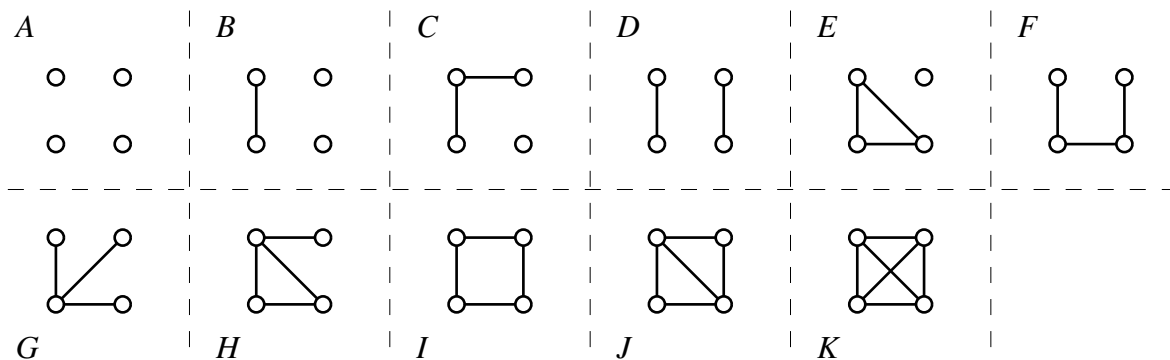
This graph, by the way, is sometimes called the *Martini Glass* or the *Alavi Graph*.

The graph below, which is an example of a *complete graph*, has order 5 and size 10. It is therefore a $(5, 10)$ graph.



Notice that we didn't give the vertices in this last graph names (like a , b , c). That's because the structure of the graph is clear from the picture, and the names that we give the vertices don't affect that structure. We'll explore this in more detail in Section 1.6.

Below are eleven graphs of order 4. The graphs $A - E$ are *disconnected graphs*, while the graphs $F - K$ are *connected*. A graph that has no edges is an *empty graph*, so graph A is an empty graph (of order 4). Graph K is a *complete graph* of order 4.



A graph of order 1 is called a *trivial graph*. A graph of order at least 2 is *nontrivial*.

1.2 Subgraphs

Let G and H be graphs. If $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$, then H is a *subgraph* of G , denoted $H \subseteq G$.

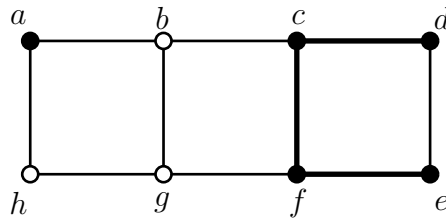
Example 1.3 Let G be the graph with

$$V(G) = \{a, b, c, d, e, f, g, h\}$$

and

$$E(G) = \{ab, bc, cd, de, ef, fg, gh, ah, bg, cf\}.$$

Let H be the graph with $V(H) = \{a, c, d, e, f\}$ and $E(H) = \{cd, cf, ef\}$. Then $H \subseteq G$. The figure below shows the graph G . The vertices of G that are also vertices of H are black, while the edges of G that are edges of H are thicker.



Let G be a graph and $S \subseteq V(G)$. The *subgraph induced by S* is the subgraph $G[S]$ having $V(G[S]) = S$ and where $uv \in E(G[S])$ if and only if $u, v \in S$ and $uv \in E(G)$.

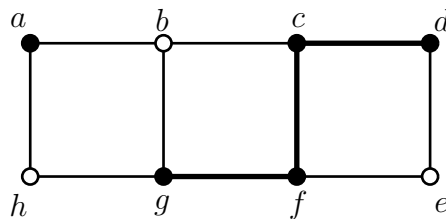
Example 1.4 Let G be the graph from Example 1.3 and let $S = \{a, c, d, f, g\}$. Then $G[S]$ is the graph with

$$V(G[S]) = \{a, c, d, f, g\}$$

and

$$E(G[S]) = \{cd, cf, fg\},$$

as indicated in the figure below:



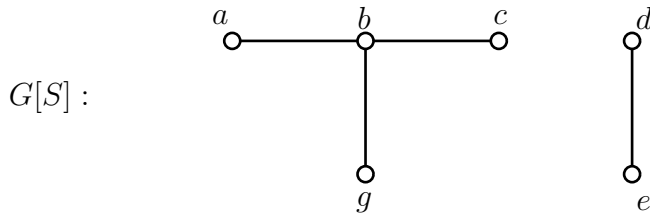
If $H \subseteq G$ and there is a set $S \subseteq V(G)$ such that $H = G[S]$, then H is called an *induced subgraph* of G .

Example 1.5 The subgraph H from Example 1.3 is not an induced subgraph of G , because although d and e are in $V(H)$ and $de \in E(G)$, there is no edge between d and e in H . The subgraph

H' with $V(H') = \{a, c, d, f, g\}$ and $E(H') = \{cd, cf, fg\}$ is an induced subgraph of G , as was demonstrated in Example 1.4.

We can define an analogous concept for edges. Let G be a graph and $S \subseteq E(G)$. Then the *subgraph induced by S* is the subgraph $G[S]$ with $E(G[S]) = S$ and where $u \in V(G[S])$ if and only if u is incident with at least one edge in S . If H is a subgraph of G and there is a set $S \subseteq E(G)$ such that $H = G[S]$, then H is called an *edge-induced subgraph* of G .

Example 1.6 Let G be the graph from Example 1.3 and let $S = \{ab, bc, bg, de\}$. Then the subgraph induced by S has vertex set $\{a, b, c, d, e, g\}$ and edge set $\{ab, bc, bg, de\}$, as shown below:



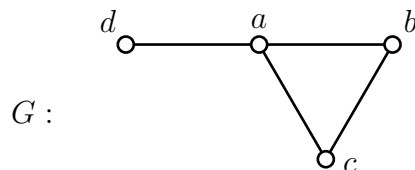
Lastly, if $H \subseteq G$ and $V(H) = V(G)$, then H is a *spanning subgraph* of G .

Example 1.7 Let G be the graph from Example 1.3. Then the graph H with $V(H) = V(G)$ and $E(H) = \{ab, bc, de, ef\}$ is a spanning subgraph of G .

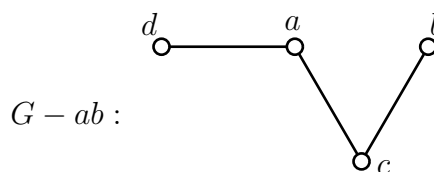
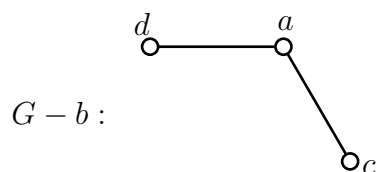
If G is a nontrivial graph and $v \in V(G)$, then by $G - v$ we mean the subgraph of G obtained from G by deleting v and every edge incident with v . Similarly, if $e \in E(G)$, then $G - e$ is the subgraph obtained from G by deleting the edge e .

Example 1.8

Consider once again the graph



The graphs $G - b$ and $G - ab$ are shown below:

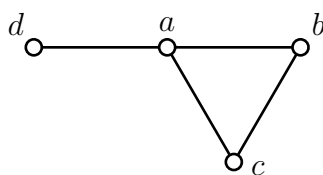


1.3 The degree of a vertex

1.3.1 Introduction

The *degree* of a vertex v is the number of edges incident with v . The degree of v is denoted $\deg v$ or, if we wish to be clear which graph we're referring to, $\deg_G v$. A vertex of degree 0 is called an *isolated vertex*. A vertex of degree 1 is called an *end-vertex*, while an edge that is incident with an end-vertex is called a *pendant edge*.

Example 1.9 Consider the graph shown below.



In this graph, we have

$$\deg a = 3$$

$$\deg b = 2$$

$$\deg c = 2$$

$$\deg d = 1$$

We are now ready to give our first (very simple) result.

Theorem 1 ('First Theorem of Graph Theory'). *If G is a graph of size m , then*

$$\sum_{v \in V(G)} \deg v = 2m.$$

Proof. When we add up the degrees of the vertices, each edge is counted exactly twice. □

The 'First Theorem of Graph Theory' has the following Corollary. A vertex of even degree is called an *even vertex*. A vertex of odd degree is an *odd vertex*.

Corollary 2. *Every graph has an even number of odd vertices.*

Proof. Let G be a graph of size m . Let V_e be the set of even vertices of G and V_o the set of odd vertices. Then $V(G) = V_e \cup V_o$ and $V_e \cap V_o = \emptyset$, so by Theorem 1 we have

$$2m = \sum_{v \in V(G)} \deg v = \sum_{v \in V_e} \deg v + \sum_{v \in V_o} \deg v,$$

so

$$\sum_{v \in V_o} \deg v = 2m - \sum_{v \in V_e} \deg v. \quad (1.1)$$

For each $v \in V_e$, we know that $\deg v$ is an even number. The right hand side of Equation (1.1) is therefore an even number. Hence $\sum_{v \in V_o} \deg v$ is an even number and therefore, by an exercise, $|V_o|$ is an even number. $\square$

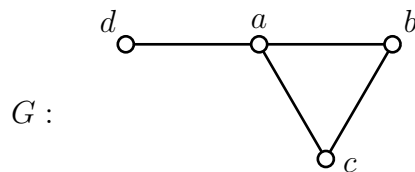
Let G be a graph. The *minimum degree* of G is

$$\delta(G) = \min\{\deg v : v \in V(G)\}$$

and the *maximum degree* of G is

$$\Delta(G) = \max\{\deg v : v \in V(G)\}.$$

Example 1.10 The graph G shown below

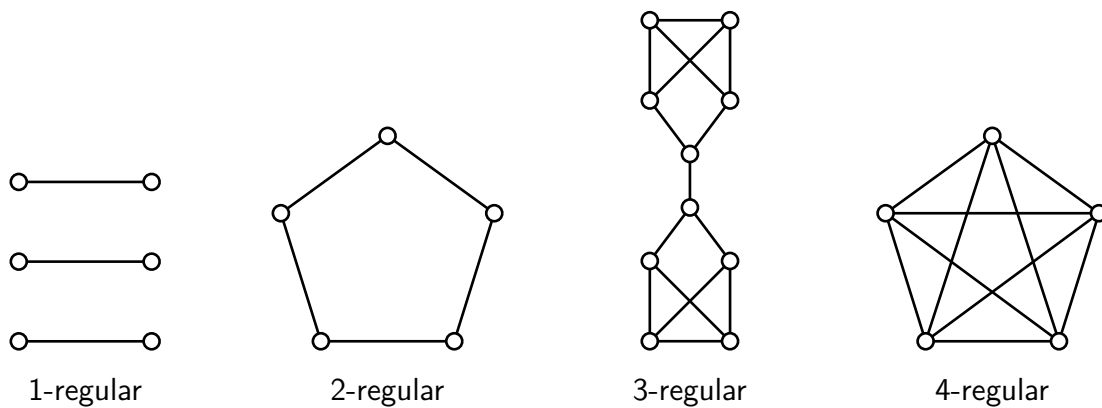


has

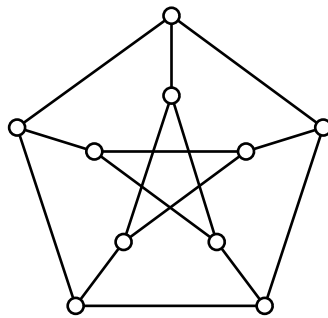
$$\begin{aligned} \delta(G) &= 1 \\ \Delta(G) &= 3. \end{aligned}$$

A graph is *regular* if every vertex has the same degree. If every vertex has degree r , then the graph is *r-regular* or *regular of degree r*.

Example 1.11 Here are some examples of regular graphs of different degrees:



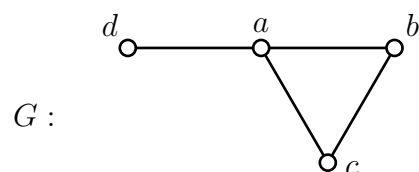
Example 1.12 A graph that is 3-regular is also called *cubic*. Probably the most famous cubic graph is the *Petersen Graph*:



1.3.2 The degree sequence of a graph

If G is a graph with vertex set $V(G) = \{v_1, v_2, \dots, v_n\}$, where $\deg v_1 \geq \deg v_2 \geq \dots \geq \deg v_n$, then $\deg v_1, \deg v_2, \dots, \deg v_n$ is the *degree sequence* of G .

Example 1.13 Consider once again the graph



Then 3, 2, 2, 1 is the degree sequence of G .

Clearly, every graph has a degree sequence. But is every finite sequence of nonnegative integers

the degree sequence of a graph?

Example 1.14 Which of the following is the degree sequence of a graph?

1. $s_1 : 4, 4, 3, 3, 3, 2, 1, 1$.
2. $s_2 : 6, 6, 3, 3, 3, 3$.
3. $s_3 : 3, 3, 3, 1$.

Solution:

1. The sequence s_1 has five odd entries. Therefore, by Corollary 2, it is not the degree sequence of a graph.
2. The sequence s_2 has six entries, so if it's the degree sequence of some graph G , then G has order 6. However, the maximum possible degree in a graph of order 6 is 5, and s_2 has two 6's. The sequence s_2 is therefore not the degree sequence of a graph.
3. Suppose that s_3 is the degree sequence of a graph G . Then G has order 4. Each of the three vertices of degree 3 is necessarily adjacent to the vertex of degree 1. However, the vertex of degree 1 is only adjacent to one other vertex. This is a contradiction and hence no such graph G exists. So s_3 is not the degree sequence of a graph.

1.3.3 The Havel-Hakimi Theorem

The preceding example shows that not every finite sequence of nonnegative integers is the degree sequence of a graph. The question of how to determine whether a given sequence of nonnegative integers is the degree sequence of a graph was first solved by Václav Havel [Hav55] in 1955. A few years later, S. Louis Hakimi [Hak62] independently discovered the same result, which now bears the names of both of these authors.

Theorem 3 (Havel-Hakimi Theorem). *A sequence $s : d_1, d_2, \dots, d_n$ of nonnegative integers with $d_1 \geq d_2 \geq \dots \geq d_n$ ($n \geq 2$ and $d_1 \geq 1$) is the degree sequence of a graph if and only if the sequence $s_1 : d_2 - 1, d_3 - 1, \dots, d_{d_1+1} - 1, d_{d_1+2}, \dots, d_n$ is the degree sequence of a graph.*

Proof. ($\Leftarrow$) Assume first that s_1 is the degree sequence of a graph G . We must prove that s is also the degree sequence of a graph. Suppose that the vertices of G are $v_2, v_3, \dots, v_n$ and that for $2 \leq i \leq d_1 + 1$ we have $\deg v_i = d_i - 1$, while for $d_1 + 2 \leq i \leq n$ we have $\deg v_i = d_i$. Let G' be the graph formed from G by adding a new vertex v_1 and joining it to the vertices $v_2, v_3, \dots, v_{d_1+1}$. Then the new graph G' has degree sequence s and hence s is the degree sequence of a graph.

($\Rightarrow$) Suppose now that there is at least one graph with degree sequence s . Amongst all such graphs, let G be one such that (i) $V(G) = \{v_1, v_2, \dots, v_n\}$, (ii) for all $i \in \{1, 2, \dots, n\}$, we have $\deg v_i = d_i$, and, (iii) the sum of the degrees of the vertices adjacent to v_1 is as large as possible. We claim that the neighbours of v_1 have degrees $d_2, d_3, \dots, d_{d_1+1}$.

Proof of claim: Suppose, by way of contradiction, that the claim is not true. Then there are vertices v_i and v_j , where $i < j$, such that (i) $d_i > d_j$, and, (ii) v_1 is adjacent to v_j but not adjacent to v_i . Since $d_i > d_j$, there is a vertex $v_{i'} \neq v_j$ that is adjacent to v_i but not to v_j . Let G' be the graph obtained from G by deleting the edges v_1v_j and $v_iv_{i'}$ and adding the edges v_1v_i and $v_jv_{i'}$. The graph G' has the same degree sequence, s , as the graph G , but in G' the sum of the degrees of the neighbours of v_1 is higher than in G . This contradicts our choice of G , so it must be the case that the neighbours of v_1 have degrees $d_2, d_3, \dots, d_{d_1+1}$, which proves the claim. $\textcircled{C}$

It follows that the graph $G - v_1$ has degree sequence s_1 , which proves that s_1 is the degree sequence of a graph. $\square$

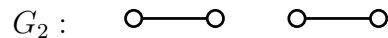
By repeated use of the Havel-Hakimi Theorem, we may now determine whether a finite sequence of nonnegative integers is the degree sequence of a graph.

Example 1.15 *Is $s : 5, 3, 3, 3, 2, 2$ the degree sequence of a graph?*

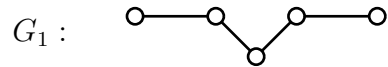
Solution: Repeatedly using the Havel-Hakimi Theorem, we have

$s : 5, 3, 3, 3, 2, 2$ is the degree sequence of a graph
if and only if $s_1 : 2, 2, 2, 1, 1$ is the degree sequence of a graph
if and only if $s_2 : 1, 1, 1, 1$ is the degree sequence of a graph.

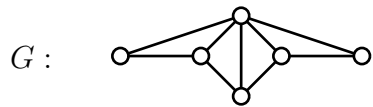
The sequence $s_2 : 1, 1, 1, 1$ is clearly the degree sequence of the graph G_2 shown here



Hence s_2 is the degree sequence of a graph, and therefore s is the degree sequence of a graph. To construct a graph with degree sequence s , we work backwards from the graph with degree sequence s_2 , using the ideas in the proof of the Havel-Hakimi Theorem. First, we construct a graph G_1 with degree sequence s_1 by adding a new vertex to G_2 and joining it to two vertices of degree 1 in G_2 :



Now we construct G from G_1 by adding another new vertex and joining it to all five vertices of G_1 :



The graph G has degree sequence $s : 5, 3, 3, 3, 2, 2$, as required.

1.3.4 The Erdős-Gallai Theorem

Another useful characterization of degree sequences of graphs was obtained by Paul Erdős and Tibor Gallai [EG60] in 1960. We give a proof only of the necessity.

Theorem 4 (Erdős-Gallai Theorem). *A sequence $d_1, d_2, \dots, d_n$ of nonnegative integers with $d_1 \geq d_2 \geq \dots \geq d_n$ ($n \geq 2$) is the degree sequence of a graph if and only if $\sum_{i=1}^n d_i$ is even and for each $k \in [1, n-1]$ we have*

$$\sum_{i=1}^k d_i \leq k(k-1) + \sum_{i=k+1}^n \min\{k, d_i\}.$$

Proof. We prove only the necessity, i.e., we prove that the degree sequence of a graph must satisfy the two conditions listed above. Let G be a nontrivial graph with vertex set $V(G) = \{v_1, v_2, \dots, v_n\}$ and degree sequence $d_1, d_2, \dots, d_n$, where $d_1 \geq d_2 \geq \dots \geq d_n$, and, for all $i \in [1, n]$, we have $\deg v_i = d_i$. We have already proved (Theorem 1) that $\sum_{i=1}^n d_i$ is even. Let $k \in [1, n-1]$ and let $V_k = \{v_1, v_2, \dots, v_k\}$. Let E_1 be the set of edges that are incident with exactly one vertex in V_k and let E_2 be the set of edges that are incident with exactly two vertices in V_k . Then $\sum_{i=1}^k d_i = |E_1| + 2|E_2|$. Certainly, $|E_2| \leq k(k-1)/2$. Furthermore, every edge in E_1 is incident with exactly one vertex in $V(G) - V_k$, and no vertex $v_i \in V(G) - V_k$ is incident with more than $\min\{k, d_i\}$ edges of E_1 . Hence $|E_1| \leq \sum_{i=k+1}^n \min\{k, d_i\}$, which proves the result. $\square$

1.4 Some classes of graphs

The *path* of order n ($n \geq 1$) is the graph P_n with $V(P_n) = \{v_1, v_2, \dots, v_n\}$ and $E(P_n) = \{v_i v_{i+1} : 1 \leq i \leq n-1\}$. Some paths are shown in Figure 1.1.

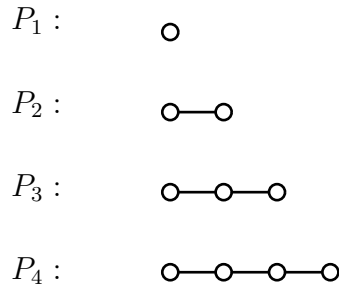


Figure 1.1: Some paths

The *cycle* of order n ($n \geq 3$) is the graph C_n with $V(C_n) = \{v_1, v_2, \dots, v_n\}$ and $E(C_n) = \{v_i v_{i+1} : 1 \leq i \leq n-1\} \cup \{v_1 v_n\}$. Some cycles are shown in Figure 1.2.

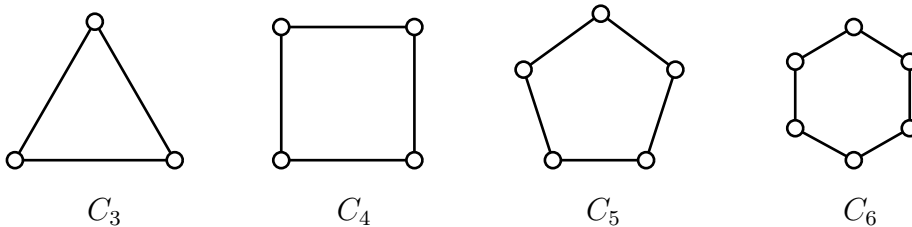


Figure 1.2: Some cycles

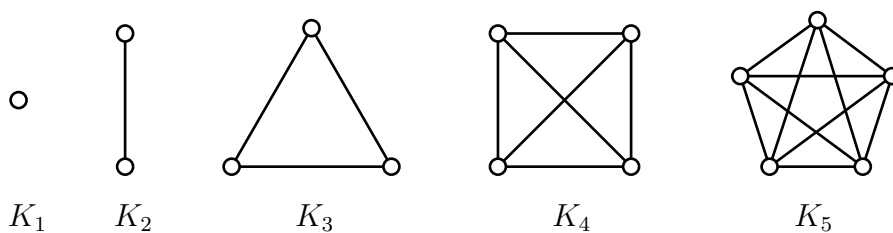


Figure 1.3: Some complete graphs

A graph is *complete* if every vertex is adjacent to every other vertex. The *complete graph of order n* ($n \geq 1$) is denoted K_n . Some complete graphs are shown in Figure 1.3.

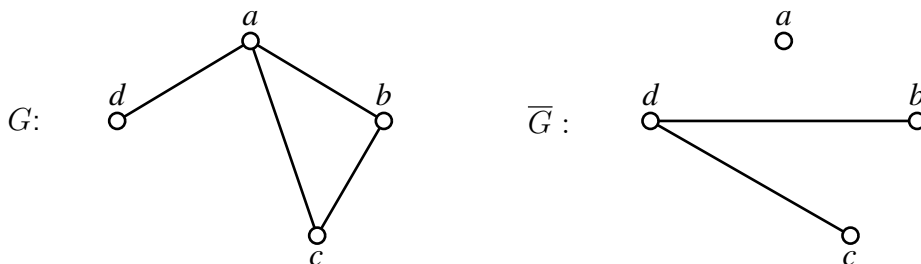
A graph in which every vertex is adjacent to every other vertex is complete. A graph in which at least one pair of vertices is not adjacent is *incomplete*.

1.5 Combining graphs

There are many different ways of generating new graphs from old. In this section, we explore some of these.

The *complement* of a graph G is the graph $\overline{G}$ with $V(\overline{G}) = V(G)$ and where two vertices are adjacent in $\overline{G}$ if and only if they are *not* adjacent in G .

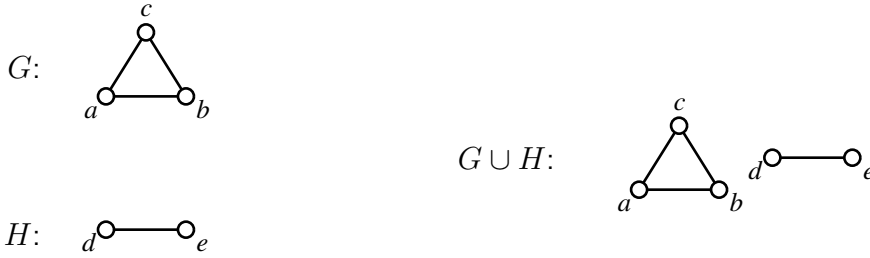
Example 1.16 A graph G and its complement $\overline{G}$ are shown below:



The *empty graph* of order n is $\overline{K_n}$, the complement of the complete graph of order n .

If G and H are graphs, then the *union* of G and H is the graph $G \cup H$ with $V(G \cup H) = V(G) \cup V(H)$ and $E(G \cup H) = E(G) \cup E(H)$. If $V(G) \cap V(H) = \emptyset$, then G and H are *disjoint*.

Example 1.17 Two disjoint graphs G and H are shown below, together with their union $G \cup H$.



If G is a graph and k a positive integer, then by kG we mean the graph consisting of k disjoint ‘copies’ of G . To define this precisely, recall that the *cartesian product of two sets* A and B is the set $A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$. We may then define kG as the graph with $V(kG) = V(G) \times [1, k]$ and $E(kG) = \{(u, i)(v, i) : uv \in E(G) \text{ and } i \in [1, k]\}$.

Example 1.18

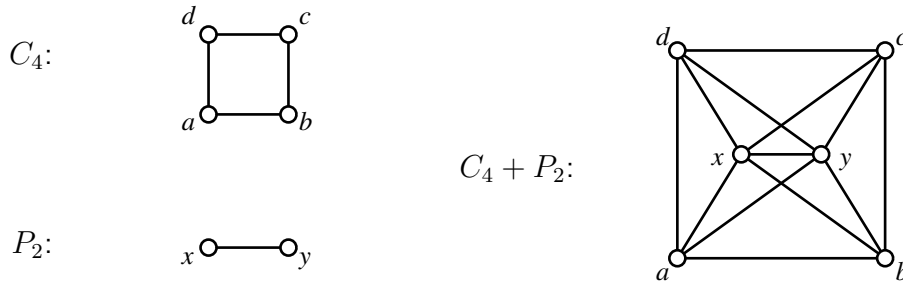
The graphs C_3 and $3C_3$ are shown below:



If G and H are disjoint graphs, then their *join* is the graph $G + H$ having $V(G + H) = V(G) \cup V(H)$ and $E(G + H) = E(G) \cup E(H) \cup \{gh : g \in V(G) \text{ and } h \in V(H)\}$.

Example 1.19

The join $C_4 + P_2$ is shown below:

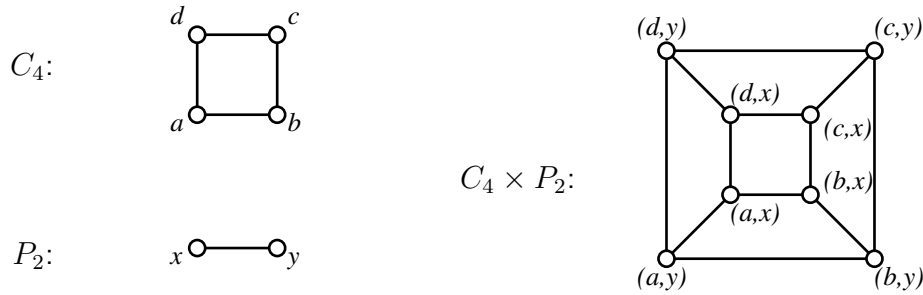


If G and H are graphs, then the *cartesian product* of G and H is the graph $G \times H$ in which

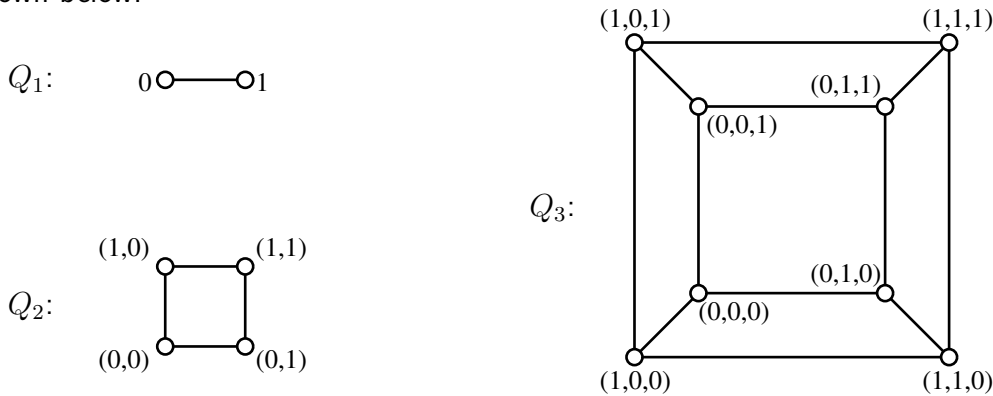
$$V(G \times H) = V(G) \times V(H), \text{ and,}$$

$$E(G \times H) = \{(g_1, h_1)(g_2, h_2) : \text{(i) } g_1g_2 \in E(G) \text{ and } h_1 = h_2, \text{ or, (ii) } g_1 = g_2 \text{ and } h_1h_2 \in E(H)\}.$$

Example 1.20 The cartesian product $C_4 \times P_2$ is shown below:



Example 1.21 In this example, we introduce an important class of graphs. We define the t -cube recursively as the graph Q_t with $Q_1 = K_2$ and, for $t \geq 2$, $Q_t = Q_{t-1} \times K_2$. The first few cubes are shown below:



The t -cube is also known as a *hypercube*.

Another way to define the cubes is as follows: For a positive integer t , the vertex set of the graph Q_t consists of all ordered binary t -tuples, and two vertices in Q_t are adjacent if the ordered binary t -tuples they represent differ in exactly one position. This is illustrated in the preceding figure.

1.6 Isomorphic graphs

Two graphs G and H are *equal* if $V(G) = V(H)$ and $E(G) = E(H)$. Consider the three graphs G_1 , G_2 , G_3 shown in Figure 1.4.

Since $V(G_1) \neq V(G_2)$, clearly G_1 and G_2 are not equal. And while $V(G_1) = V(G_3)$, notice that $ac \in E(G_1)$ but $ac \notin E(G_3)$, so $E(G_1) \neq E(G_3)$, which means that $G_1 \neq G_3$.

It should be obvious, though, that these three graphs are, at a structural level, the same. The way we formalize this idea is as follows: If G and H are graphs, then an *isomorphism* from

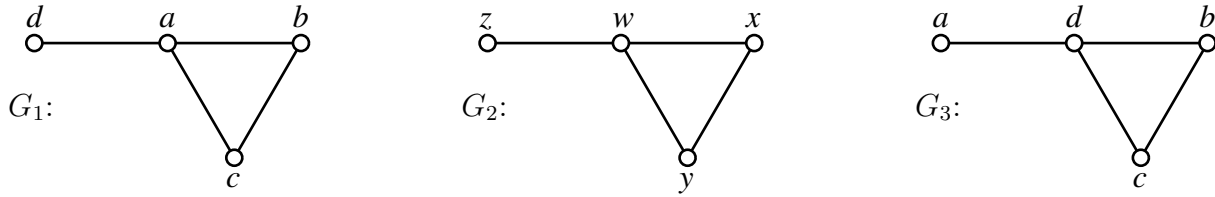
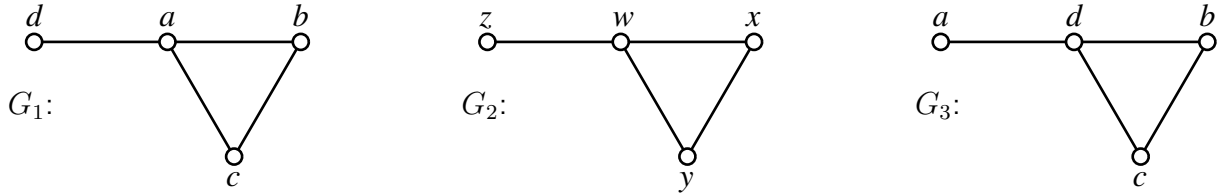


Figure 1.4: Three pairwise unequal but isomorphic graphs.

G to H is a bijection $f : V(G) \rightarrow V(H)$ such that for all $u, v \in V(G)$, we have $uv \in E(G)$ if and only if $f(u)f(v) \in E(H)$. If there is an isomorphism between G and H , then we say that G and H are *isomorphic* graphs, and we write $G \cong H$.

Example 1.22 Consider once again the three graphs G_1, G_2, G_3 discussed previously:



Let $f_{12} : V(G_1) \rightarrow V(G_2)$ be the function defined by

$$\begin{aligned} a &\mapsto w \\ b &\mapsto x \\ c &\mapsto y \\ d &\mapsto z \end{aligned}$$

Clearly, f_{12} is a bijection. We can also check that for all $u, v \in V(G_1)$ we have $uv \in E(G_1)$ if and only if $f_{12}(u)f_{12}(v) \in E(G_2)$ (e.g., $ad \in E(G_1)$, and $f_{12}(a)f_{12}(d) = wz \in E(G_2)$). So f_{12} is an isomorphism between G_1 and G_2 , which means that $G_1 \cong G_2$, i.e., G_1 and G_2 are really the same graph with different labels on the vertices.

Notice that f_{12} is not the only isomorphism between G_1 and G_2 : The function $f'_{12} : V(G_1) \rightarrow V(G_2)$ with rule

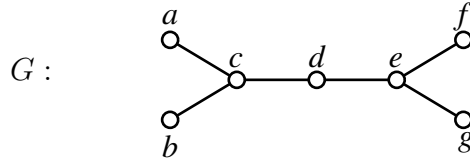
$$\begin{aligned} a &\mapsto w \\ b &\mapsto y \\ c &\mapsto x \\ d &\mapsto z \end{aligned}$$

is also an isomorphism.

Similarly, one may find isomorphisms between G_1 and G_3 and between G_2 and G_3 , showing that $G_1 \cong G_2 \cong G_3$.

Let G be a graph. An isomorphism from G to itself is called an *automorphism* of G . An automorphism of G is a permutation of $V(G)$ that preserves adjacency. We shall therefore use *cycle notation* to describe an automorphism.

Example 1.23 Consider the graph G shown below:



One automorphism of G is the permutation α_1 with rule

$$a \mapsto b$$

$$b \mapsto a$$

$$c \mapsto c$$

$$d \mapsto d$$

$$e \mapsto e$$

$$f \mapsto f$$

$$g \mapsto g$$

This may be written more succinctly using cycle notation as $\alpha_1 = (ab)(c)(d)(e)(f)(g)$, or, if we agree that any vertex that does not appear is fixed (e.g., $\alpha_1(c) = c$), we may simply write $\alpha_1 = (ab)$. With this convention, we find that there are eight automorphisms of G :

$$\alpha_e = (a) \quad (\text{the identity permutation})$$

$$\alpha_1 = (ab)$$

$$\alpha_2 = (fg)$$

$$\alpha_3 = (ab)(fg)$$

$$\alpha_4 = (af)(ce)(bg)$$

$$\alpha_5 = (afbg)(ce)$$

$$\alpha_6 = (ag)(ce)(bf)$$

$$\alpha_7 = (agbf)(ce)$$

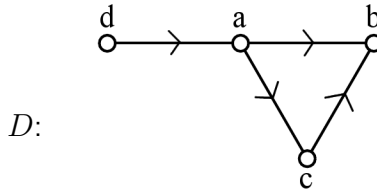
Each automorphism represents a *symmetry* of the graph: A way in which the bits of the graph may ‘rearrange’ themselves without changing the graph’s structure. We shall have more to say about this later.

1.7 Digraphs

A *directed graph* or *digraph* D consists of a finite set $V(D)$ of vertices and a set $A(D)$ of ordered pairs of distinct vertices called *arcs*. The number of vertices in a digraph is its *order*. The

number of arcs is its *size*. When we draw a picture of a digraph, we put arrows on the arcs to indicate order, e.g., the arc (a, b) is drawn with the arrow pointing from a to b . We shall denote the arc (a, b) as $\vec{ab}$.

Example 1.24 Let D be the digraph with vertex set $V(D) = \{a, b, c, d\}$ and arc set $A(D) = \{\vec{ab}, \vec{ac}, \vec{cb}, \vec{da}\}$. A diagram of D is shown below:



The digraph D has order 4 and size 4. Since $\vec{ab}$ is an arc of D , we say that a is *adjacent to* b , while b is *adjacent from* a .

Let D be a digraph and $v \in V(D)$. The *indegree* of v is the number of vertices from which v is adjacent, while the *outdegree* of v is the number of vertices to which v is adjacent. The indegree of v is denoted $\text{id } v$, while the outdegree of v is $\text{od } v$.

Example 1.25 In the digraph D in Example 1.24, we have $\text{id } a = 1$ and $\text{od } a = 2$. A vertex with outdegree 0, like the vertex b , is called a *sink*. A vertex of indegree 0, like the vertex d , is called a *source*.

If D is a digraph, then the *underlying graph* of D is the graph G obtained from D by replacing each (directed) arc with an (undirected) edge and then deleting any multiple edges which result. In other words, the underlying graph G has $V(G) = V(D)$ and $E(G) = \{uv : \vec{uv} \in A(D) \text{ or } \vec{vu} \in A(D)\}$.

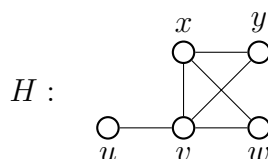
Example 1.26 The underlying graph of the digraph in Example 1.24 is the frequently discussed Martini Glass (Example 1.1).

Exercises

- 1.1 Draw the graph G with vertex set $V(G) = \{a, b, c, d, e, f\}$ and edge set $E(G) = \{ab, bc, be, cf, de, af\}$.
- 1.2 Determine the order and size of each of the following:
 - a. The path P_n .
 - b. The cycle C_n .
 - c. The complete graph K_n .
 - d. The cube Q_t .
- 1.3 Let $c_1, c_2, \dots, c_k$ be odd integers. Prove that $\sum_{i=1}^k c_i$ is odd if and only if k is odd.
- 1.4 Find a cubic graph of order 7 or prove that no such graph exists.
- 1.5 A graph G is *irregular* if no two vertices of G have the same degree. Prove that if G is an irregular graph, then G is trivial.
- 1.6 Suppose that you and your wife attend a party with three other married couples. Several handshakes take place. No-one shakes hands with himself (or herself) or with his (or her) spouse, and no-one shakes hands with the same person more than once. After all the handshaking is complete, you ask each person, including your wife, how many hands he or she shook. Each person gives a different answer.
 - a. How many hands did you shake?
 - b. How many hands did your wife shake?
- 1.7 Prove or disprove: For every graph G and every integer $r \geq \Delta(G)$, there is an r -regular graph H containing G as an induced subgraph.
- 1.8 Prove or disprove:
 - a. If two graphs are isomorphic, then they have the same degree sequence.
 - b. If two graphs have the same degree sequence, then they are isomorphic.
- 1.9 Determine whether each of the following sequences is the degree sequence of a graph. If it isn't, state why it isn't. If it is, construct a graph with that degree sequence. It is suggested, though not compulsory, that you apply both the Havel-Hakimi and Erdős-Gallai Theorems to each sequence.
 - a. 5, 5, 5, 3, 3, 2, 2, 2, 2, 2.
 - b. 4, 4, 3, 2, 1, 0.
 - c. 3, 3, 2, 2, 2, 2, 1, 1.
 - d. 7, 4, 3, 3, 2, 2, 2, 1, 1, 1.

- 1.10 Recall that an *equivalence relation* is a relation that is reflexive, symmetric, and transitive. Prove that $\cong$ is an equivalence relation on the set of all graphs. Of what significance are the equivalence classes in this relation?
- 1.11 We say that a word W_1 can be *transformed* into a word W_2 if W_2 can be obtained from W_1 by performing exactly one of the following two steps:
- Interchanging two letters of W_1 ;
 - Replacing a letter in W_1 by another letter.
- a. Let Ω be a set of words and R the relation on Ω given by: W_1RW_2 if W_1 can be transformed into W_2 . Which properties does this relation have? Is it an equivalence relation?
- b. This situation can be modeled by a graph G_Ω in which V is the set of words and we join two vertices u and v with an edge if the word represented by u can be transformed into the word represented by v . Draw the graph G_Ω for the case when $\Omega = \{\text{aim, arm, arc, art, car, oar, act, cat, rat, tar, oat}\}$.
- 1.12 How many automorphisms does each of the following graphs have?
- a. The path P_n .
 - b. The cycle C_n .
 - c. The complete graph K_n .
 - d. The empty graph $\overline{K}_n$.
- 1.13 Recall that the ‘First Theorem of Graph Theory’ (Theorem 1) states that if G is a graph of size m , then
- $$\sum_{v \in V(G)} \deg v = 2m.$$
- State and prove a ‘First Theorem of Digraph Theory’.
- 1.14 Construct the digraph with vertex set $\{-3, 3, 6, 12\}$ in which $\vec{ij}$ is an arc iff $i \neq j$ and $i|j$.
- 1.15 For the following pairs G, H of graphs, draw $G + H$ and $G \times H$:
- a. $G = K_5$ and $H = K_2$.
 - b. $G = \overline{K}_5$ and $H = \overline{K}_3$.
 - c. $G = C_5$ and $H = K_1$.
- 1.16 Find spanning subgraphs G_0, G_1, G_2, G_3 of the Petersen graph, where G_r is r -regular for $0 \leq r \leq 3$.
- 1.17 Prove that if G and $\overline{G}$ are both r -regular for some nonnegative integer r , then G has odd order.
- 1.18 A graph is *self-complementary* if it is isomorphic to its complement.

- a. Determine all self-complementary graphs of order 5 or less.
- b. If G is a self-complementary graph of order n , prove that $n \equiv 0 \pmod{4}$ or $n \equiv 1 \pmod{4}$.
- 1.19 Let n, k, r be positive integers with $r \leq k \leq n$ and let S be a set of cardinality n . Then $J(n, k, r)$ is the graph whose vertices are the k -subsets of S , and where two vertices are adjacent if the intersection of the two subsets to which they correspond has cardinality r .
- a. Draw $J(5, 2, 0)$.
- b. Determine the order of $J(n, k, r)$.
- c. Prove that $J(n, k, r)$ is regular and determine the degree of the vertices. Hence find the size of $J(n, k, r)$.
- d. Prove that $J(5, 2, 0)$ is (isomorphic to) the Petersen Graph.
- 1.20 For each of the properties listed below, find a graph G that has the given property:
- a. G has order 9 and size 6.
- b. Every vertex of G is adjacent to two vertices, every edge of G is adjacent to two edges, and G has order 10.
- c. Every two vertices of G are adjacent and every two edges of G are adjacent.
- d. G has positive size and every vertex of G is incident with every edge of G .
- e. Every vertex of G is incident with an edge of G , but no two edges are adjacent.
- f. G has order 10 and size 13 and contains the graph H below as an induced subgraph:



- 1.21 State whether each of the following statements is True or False. If True, give a proof. If False, provide a counterexample.
- a. Every subgraph of a complete graph is complete.
- b. Every induced subgraph of a complete graph is complete.
- c. Every induced subgraph of a cycle is a cycle.
- d. Every induced subgraph of a path is a path.

Chapter 2

Distance

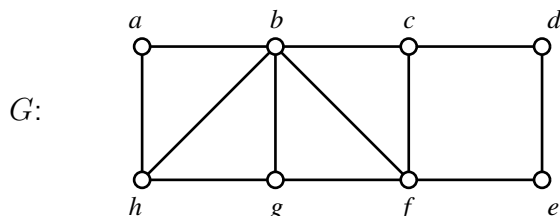
2.1 Walks, trails, paths, circuits, and cycles

Let u and v be vertices in a graph G . A u - v *walk* is a finite, alternating sequence

$$u = v_0, e_1, v_1, e_2, v_2, \dots, e_k, v_k = v$$

of vertices $v_0, v_1, \dots, v_k$ and edges $e_1, e_2, \dots, e_k$ such that, for all $i \in [1, k]$, the edge e_i joins the vertices v_{i-1} and v_i . The number of edges in a walk is its *length*, and a walk of length zero is *trivial*.

Example 2.1 Consider the graph G shown below:



The sequence $W_1 : a, ab, b, bc, c, bc, b, bg, g, fg, f$ is an $a - f$ walk of length 5.

For brevity, we shall usually write walks by listing the vertices but omitting the edges. For example, the walk W_1 in Example 2.1 could be rewritten as $W_1 : a, b, c, b, g, f$.

A *trail* is a walk in which no edge is repeated. A *path* is a walk in which no vertex is repeated. A walk that begins and ends at the same vertex is *closed*. A walk that is not closed is *open*. A nontrivial closed trail is called a *circuit*. A circuit with no repeated vertex except the first/last is a *cycle*.

Example 2.2

Consider again the graph G from Example 2.1. The walk W_1 from that example is neither a trail nor a path, since it contains repeated vertices and edges. The walk $W_2 : c, f, g, b, c, d$ is a trail, since no edge is repeated, but not a path, since the vertex c is repeated. The walk $W_3 : h, g, b, c, f$

is a path (and a trail). The walks W_1 , W_2 , and W_3 are all open, while the walk $W_4 : a, b, a, b, a, b, a$ is closed. The walk $W_5 : b, a, h, b, f, c, b$ is a circuit and the walk $W_6 : e, d, c, b, f, e$ is a cycle.

Every $u - v$ path is a $u - v$ trail, and every $u - v$ trail is (of course) a $u - v$ walk. It is not true, however, that every $u - v$ walk is a $u - v$ path. A walk W is said to *contain* a walk W' if W' is a subsequence of W .

Theorem 5. *Every $u - v$ walk in a graph contains a $u - v$ path.*

Proof. Let W be a $u - v$ walk in a graph G . If W is closed, then $W' : u$ is a (trivial) $u - v$ path and we are done. Suppose then that $u \neq v$. If no vertex of W is repeated, then W is a $u - v$ path, so we may suppose that at least one vertex x of G occurs at least twice on W . Suppose that $W : v_0, v_1, \dots, v_k$ and let i and j , with $0 \leq i < j \leq k$, be the smallest and largest integers respectively such that $x = v_i = v_j$. Then $W' : v_0, v_1, \dots, v_i, v_{j+1}, v_{j+2}, \dots, v_k$ is a $u - v$ walk in which the vertex x is not repeated. If W' contains no repeated vertex, then W' is a $u - v$ path contained in W . If W' contains a repeated vertex, then, since W is finite, we may repeat the above procedure as many times as necessary until we obtain a $u - v$ path that is contained in W . $\square$

2.2 Distance in graphs

Let u and v be vertices in a graph G . We say that u is *connected to* v if there is a $u - v$ path in G . A graph G is *connected* if every pair of vertices of G is connected. A graph that is not connected is *disconnected*. A maximal connected subgraph¹ of G is called a *component* of G .

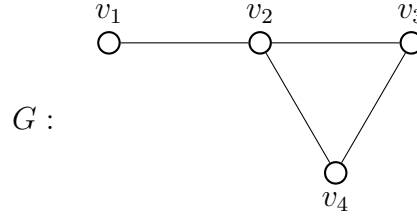
Example 2.3



The graph G_1 is connected. The graph G_2 , which has four components, is disconnected.

Let G be a connected graph and $u, v \in V(G)$. The *distance* from u to v , denoted $d(u, v)$, is the minimum length of a $u - v$ path. A $u - v$ path of minimum length is called a *shortest* $u - v$ path or a $u - v$ *geodesic*.

¹A subgraph is *maximal* with respect to some property if it is not properly contained in any other subgraph with the same property. Hence, a maximal *connected* subgraph is a connected subgraph that is not properly contained in any other *connected* subgraph.

Example 2.4

In the graph G above, we have $d(v_1, v_1) = 0$, while $d(v_1, v_2) = 1$ and $d(v_1, v_3) = d(v_1, v_4) = 2$. The path v_1, v_2, v_3 is a $v_1 - v_3$ geodesic. The path v_1, v_2, v_4, v_3 is a $v_1 - v_3$ path but not a $v_1 - v_3$ geodesic.

Recall that a *metric space* is an ordered pair (S, f) consisting of a set S and a function $f : S \times S \rightarrow \mathbb{R}$ satisfying for all $x, y, z \in S$:

1. $f(x, y) \geq 0$,
2. $f(x, y) = 0$ if and only if $x = y$,
3. $f(x, y) = f(y, x)$, and,
4. $f(x, y) \leq f(x, z) + f(z, y)$ (the triangle inequality).

Theorem 6. *If G is a connected graph and $d : V(G) \times V(G) \rightarrow \mathbb{Z}$ is the standard distance function, then $(V(G), d)$ is a metric space.*

The proof of this is left as an exercise.

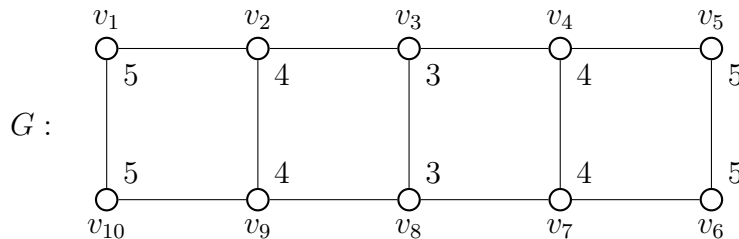
Let G be a connected graph and u a vertex of G . The *eccentricity* of the vertex u is

$$e(u) = \max\{d(u, v) : v \in V(G)\}.$$

The *radius* and *diameter* of the graph G are then defined as

$$\begin{aligned} \text{rad } G &= \min\{e(u) : u \in V(G)\} \\ \text{diam } G &= \max\{e(u) : u \in V(G)\}. \end{aligned}$$

If $e(u) = \text{rad } G$, then u is a *central vertex* of G , while if $e(u) = \text{diam } G$, then u is a *peripheral vertex* of G . The subgraph of G induced by the central vertices of G is the *center* of G , while the subgraph induced by the peripheral vertices is the *periphery*.

Example 2.5

In the graph G shown above, each vertex is labeled with its eccentricity. We have $\text{rad } G = 3$ and $\text{diam } G = 5$. The central vertices of G are v_3 and v_8 , and the center of G , the subgraph induced by $\{v_3, v_8\}$, is isomorphic to K_2 . The peripheral vertices of G are v_1 , v_5 , v_6 , and v_{10} , and the periphery of G , the subgraph induced by $\{v_1, v_5, v_6, v_{10}\}$, is isomorphic to $2K_2$.

Theorem 7. For every connected graph G ,

$$\text{rad } G \leq \text{diam } G \leq 2 \text{ rad } G.$$

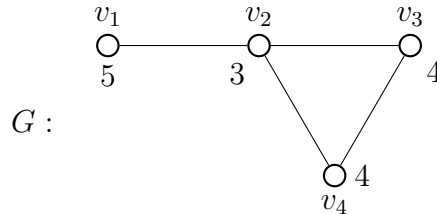
Proof. The first part of the chain of inequalities, that $\text{rad } G \leq \text{diam } G$, follows directly from their definition. To prove that $\text{diam } G \leq 2 \text{ rad } G$, let p_1 and p_2 be vertices of G for which $d(p_1, p_2) = \text{diam } G$, and let c be a central vertex of G . Using the triangle inequality, we have $\text{diam } G = d(p_1, p_2) \leq d(p_1, c) + d(c, p_2) \leq e(c) + e(c) = 2 \text{ rad } G$. $\square$

For a vertex u in a connected graph G , the *total distance* of u is

$$d(u) = \sum_{v \in V(G)} d(u, v).$$

If $d(u) = \min\{d(v) : v \in V(G)\}$, then v is a *median vertex* of G . The subgraph of G induced by the median vertices is the *median* of G .

Example 2.6



Each vertex of the graph G shown above is labeled with its total distance. The only median vertex is v_2 . The median of G consists of the graph induced by $\{v_2\}$, which is isomorphic to K_1 .

2.3 Walks, connectivity, and distance in digraphs

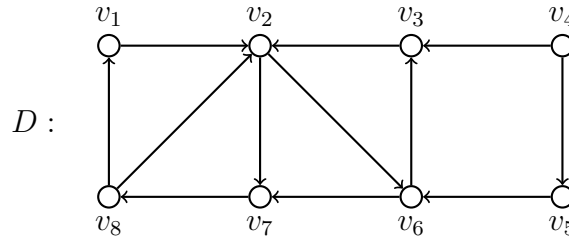
When discussing digraphs, we must be careful to distinguish between those situations in which we wish to take into account the directions of the edges, and those in which we do not.

Let D be a digraph and u, v vertices of D .

- A $u - v$ *directed walk* is a finite alternating sequence $u = v_0, a_1, v_1, a_2, v_2, \dots, a_k, v_k = v$ of vertices and arcs of D in which, for all $i \in [1, k]$, we have $a_i = \overrightarrow{v_{i-1}v_i}$.
- A $u - v$ *semi-walk* is a finite alternating sequence $u = v_0, a_1, v_1, a_2, v_2, \dots, a_k, v_k = v$ of vertices and arcs of D in which, for all $i \in [1, k]$, we have $a_i = \overrightarrow{v_{i-1}v_i}$ or $a_i = \overrightarrow{v_i v_{i-1}}$.

To summarize: In a directed walk, we must respect the directions of the arcs, while in a semi-walk, we need not.

Example 2.7



In the digraph D shown above, v_1, v_2, v_6, v_3, v_2 is both a $v_1 - v_2$ directed walk and a $v_1 - v_2$ semi-walk. On the other hand, v_1, v_8, v_2 is a $v_1 - v_2$ semi-walk, but not a $v_1 - v_2$ directed walk.

Analogously with undirected graphs, a $u - v$ *directed path* is a $u - v$ directed walk in which no vertex is repeated, while a $u - v$ *semi-path* is a $u - v$ semi-walk in which no vertex is repeated.

A digraph D is called *connected* or *weakly connected* if for every pair u, v of vertices of D there is a $u - v$ semi-path, i.e., if the underlying graph of D is connected. On the other hand, D is *strongly connected* if for every pair u, v of vertices there is a $u - v$ directed path.

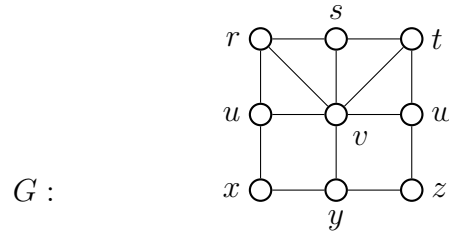
Example 2.8 The digraph in Example 2.7 is weakly connected (since the underlying graph is connected). It is not, however, strongly connected (there is no $v_1 - v_4$ directed path in D).

If D is a connected digraph and $u, v \in V(D)$, then $d(u, v)$, the *distance between u and v* , is the smallest number of arcs on a $u - v$ directed path. If D is not strongly connected, then there will be pairs u, v of vertices for which $d(u, v)$ is not well-defined; for convenience, we frequently define the distance between such pairs to be ∞ .

Example 2.9 Returning to the digraph in Example 2.7, we have $d(v_1, v_8) = 3$ while $d(v_1, v_4) = \infty$.

Exercises

2.1 Consider the following graph:



Give an example of each of the following or explain why no such example exists:

- a. An $x - y$ walk of length 6.
- b. A $v - w$ trail that is not a $v - w$ path.
- c. An $r - z$ path of length 2.
- d. An $x - z$ path of length 3.
- e. An $x - t$ path of length $d(x, t)$.
- f. A circuit of length 10.
- g. A cycle of length 8.
- h. A geodesic whose length is $\text{diam } G$.

2.2 Prove that a closed walk of odd length contains an odd cycle.

2.3 Recall that an *equivalence relation* is a relation that is reflexive, symmetric, and transitive. If G is a graph, prove that *is connected to* is an equivalence relation on $V(G)$. Of what significance are the equivalence classes in this relation?

2.4 Prove Theorem 6.

2.5 If G is a graph of order n with $\delta(G) \geq \frac{n-1}{2}$, prove that G is connected.

2.6 Prove or disprove: If G is a connected graph and $v \in V(G)$, then v is a central vertex if and only if v is a median vertex.

2.7 Prove that if G is a disconnected graph, then $\text{diam } \overline{G} \leq 2$.

2.8 Prove or disprove: Every graph is the center of some connected graph.

2.9 For a connected graph G , the *average distance* $\mu(G)$ of the graph is the average distance between all pairs of vertices, i.e., $\mu(G) = \frac{2}{n(n-1)} \sum_{u,v \in V(G)} d(u, v)$. Prove or disprove: If G is a connected graph, then $\text{rad } G \leq \mu(G) \leq \text{diam } G$.

2.10 Draw all connected graphs of order 5 in which the distance between every two distinct vertices is odd. Explain why you know that you have drawn all such graphs.

- 2.11 Let $P : u = v_0, v_1, \dots, v_k = v$ be a $u - v$ geodesic in a connected graph G . Prove that $d(u, v_i) = i$ for each integer i with $1 \leq i \leq k$.
- 2.12 Let P and Q be two longest paths in a connected graph G . Prove that P and Q have at least one vertex in common.
- 2.13 Prove or disprove: Let G be a connected graph of diameter k . If P and Q are two geodesics of length k in G , then P and Q have at least one vertex in common.
- 2.14 Show that if G is a disconnected graph containing exactly two odd vertices, then these odd vertices must be in the same component of G .
- 2.15 Show that if G is a connected graph that is not regular, then G contains adjacent vertices u and v such that $\deg u \neq \deg v$.
- 2.16 A graph is *self-complementary* if it is isomorphic to its complement. Prove that every nontrivial self-complementary graph has diameter 2 or 3.
- 2.17 If u and v are adjacent vertices in a connected graph G , prove that $|e(u) - e(v)| \leq 1$.
- 2.18 Let D be a digraph. For a vertex v in D , the quantity $e^+(v) = \max\{d(v, x) : x \in V(D)\}$ is called the *out-eccentricity* of v .
- Prove or disprove: Every vertex in D has finite out-eccentricity if and only if D is strongly connected.
 - Let u and v be vertices in D . Prove or disprove: If D is strongly connected and $\vec{uv} \in A(D)$, then $|e^+(u) - e^+(v)| \leq 1$.
- 2.19 Let G be a connected graph and u a vertex of G . A vertex u' is called an *eccentric vertex* of u if $d(u, u') = e(u)$.
- If c is a central vertex of G and u' an eccentric vertex of u and $e(u) > \text{rad } G$, prove that u does not lie between c and u' (i.e., there is no $c - u'$ geodesic containing u).
 - Hence prove that if k is some integer satisfying $\text{rad } G < k \leq \text{diam } G$, then G contains (at least) two vertices of eccentricity k .

Chapter 3

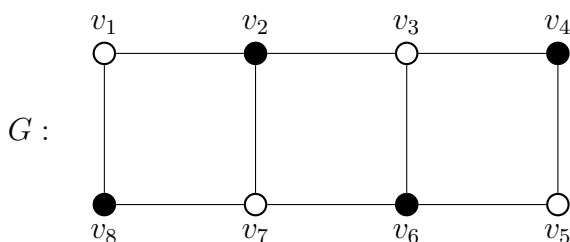
The structure of graphs

3.1 Bipartite graphs

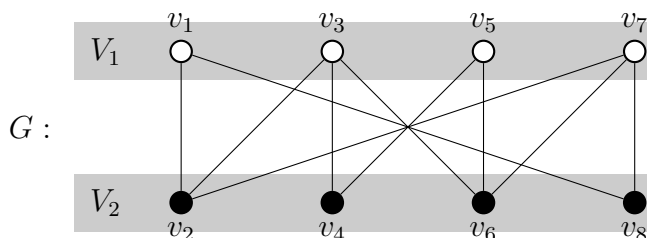
Recall that a collection $\{S_1, S_2, \dots, S_k\}$ of sets is called *pairwise disjoint* if for all $i, j \in [1, k]$ with $i \neq j$, we have $S_i \cap S_j = \emptyset$. A *partition* of a set S is a finite, pairwise disjoint collection $\{S_1, S_2, \dots, S_k\}$ of nonempty subsets of S such that $\bigcup_{i=1}^k S_i = S$.

A graph G is *bipartite* if there is a partition $\{V_1, V_2\}$ of $V(G)$ such that every edge of G joins a vertex of V_1 to a vertex of V_2 . In other words, G is bipartite if we can partition $V(G)$ into two sets V_1 and V_2 such that both $G[V_1]$ and $G[V_2]$ are empty. The sets V_1 and V_2 are called *partite sets* and $\{V_1, V_2\}$ is called a *bipartition* of G .

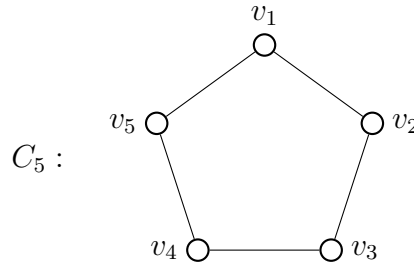
Example 3.1



In the graph G shown above, choose $V_1 = \{v_1, v_3, v_5, v_7\}$ (the white vertices) and $V_2 = \{v_2, v_4, v_6, v_8\}$ (the black vertices). Then $\{V_1, V_2\}$ is a partition of $V(G)$. Furthermore, there is no pair of adjacent white vertices, and there is no pair of adjacent black vertices. This shows that G is bipartite. To make things clearer, we can redraw G :



Not all graphs are bipartite. Consider the graph C_5 , shown below:



Suppose that C_5 is bipartite and let $\{V_1, V_2\}$ be a bipartition of C_5 . Without loss of generality, we may assume that $v_1 \in V_1$. Since $v_1v_2 \in E(C_5)$, we must then have $v_2 \in V_2$. Continuing in this fashion, we argue that $v_3 \in V_1$ and $v_4 \in V_2$. What do we do with v_5 ? We can't have $v_5 \in V_1$, because $v_1 \in V_1$ and $v_1v_5 \in E(C_5)$. Similarly, we can't have $v_5 \in V_2$ since $v_4 \in V_2$ and $v_4v_5 \in E(C_5)$. Hence C_5 is not bipartite.

With a little thought, you should convince yourself that the problem we encountered when trying to construct a bipartition of C_5 will arise with any odd cycle. The following result shows that this is an important observation. If $P : v_1, v_2, \dots, v_k$ is a path, then by $P[v_i, v_j]$ (where $i, j \in [1, k]$ and $i \leq j$) we mean the subpath of P induced by $\{v_i, v_{i+1}, \dots, v_j\}$.

Theorem 8. *A graph is bipartite if and only if it does not contain an odd cycle.*

Proof. ($\implies$) Suppose, to the contrary, that G is a bipartite graph that contains an odd cycle $v_1, v_2, \dots, v_k, v_1$. Let $\{V_1, V_2\}$ be a bipartition of G and suppose, without loss of generality, that $v_1 \in V_1$. Arguing as in Example 3.1, we find that $v_i \in V_1$ when i is odd, while $v_i \in V_2$ when i is even. But then both the vertices v_1 and v_k are in V_1 . Since $v_1v_k \in E(G)$, this is impossible.

($\impliedby$) Let G be a graph that contains no odd cycle. Notice that G is bipartite if and only if each of its components is bipartite, so we assume that G is connected. Let $u \in V(G)$ and define

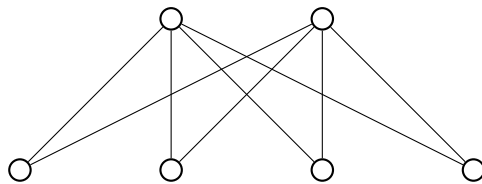
$$V_1 = \{v \in V(G) : d(u, v) \text{ is odd}\},$$

$$V_2 = \{v \in V(G) : d(u, v) \text{ is even}\}.$$

We claim that $\{V_1, V_2\}$ is a bipartition of G . Certainly, $\{V_1, V_2\}$ is a partition of $V(G)$. It remains to prove that every edge of G joins a vertex of V_1 to a vertex of V_2 . Suppose, by way of contradiction, that there are two vertices $x, y \in V_1$, say, such that $xy \in E(G)$. Let P_x be a $u - x$ geodesic and let P_y be a $u - y$ geodesic. Let P be the closed walk that starts at u , follows P_x to x , then the edge xy , then the path P_y in reverse from y to u . The closed walk P has length $d(u, x) + d(u, y) + 1$, which is odd. Hence, by a homework question, P contains an odd cycle, a contradiction, and thus $\{V_1, V_2\}$ is a bipartition of G . □

For positive integers r and s , we define the *complete bipartite graph* $K_{r,s}$ to be the graph $\overline{K_r} + \overline{K_s}$. In other words, $K_{r,s}$ is a bipartite graph in which there is a partite set V_1 of cardinality r and a partite set V_2 of cardinality s , and in which every vertex in V_1 is adjacent to every vertex in V_2 . The complete bipartite graph $K_{2,4}$ is shown in Figure 3.1. A complete bipartite graph of the form $K_{1,r}$ is called a *star*. The star $K_{1,3}$ is called a *claw*.

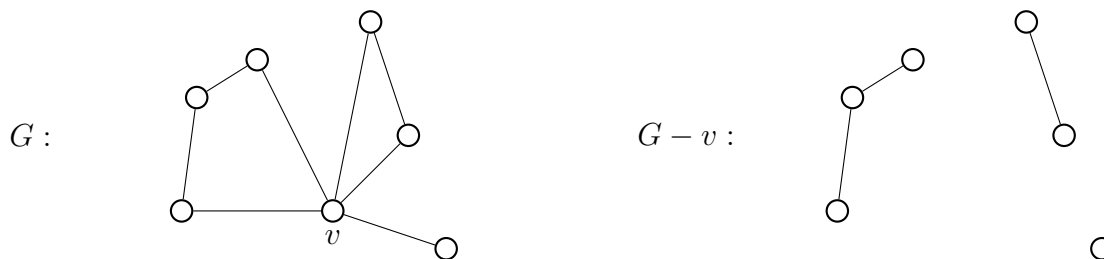
We can extend the idea of a bipartite graph. If there is a partition $\{V_1, V_2, \dots, V_k\}$ of a graph G such that for all $i \in [1, k]$, the subgraph $G[V_i]$ is empty, then G is called *k-partite*.

Figure 3.1: The complete bipartite graph $K_{2,4}$.

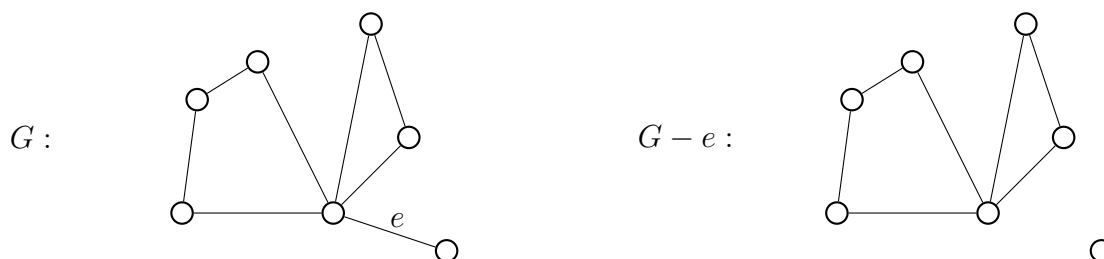
3.2 Cut-vertices, bridges, and blocks

Let G be a connected graph. If v is a vertex of G for which $G - v$ is disconnected, then v is called a *cut-vertex*. If e is an edge of G for which $G - e$ is disconnected, then e is called a *bridge*.

Example 3.2



The graph G shown above is connected while the graph $G - v$ is disconnected, so v is a cut-vertex. In fact, v is the only cut-vertex of G .



Since G is connected and $G - e$ is disconnected, e is a bridge. The edge e is the only bridge of G .

Theorem 9. A vertex v in a connected graph G is a cut-vertex if and only if there are vertices u and w ($u, w \neq v$) such that v lies on every $u - w$ path.

Proof. ($\implies$) Suppose first that v is a cut-vertex of G . Then $G - v$ is disconnected, which means that there are vertices u and w in $G - v$ such that there is no $u - w$ path in $G - v$. Hence there is no $u - w$ path in G that does not contain v .

($\impliedby$) Suppose now that there are vertices u and w with $u, w \neq v$ such that v lies on every $u - w$ path in G . Then there is no $u - w$ path in $G - v$, which means that $G - v$ is disconnected. Hence v is a cut-vertex. $\square$

An analogous result holds for bridges.

Theorem 10. *An edge e in a connected graph G is a bridge if and only if there are vertices u and w such that e lies on every $u - w$ path.*

Another interesting result for bridges is the following.

Theorem 11. *An edge e in a connected graph G is a bridge if and only if e does not lie on a cycle of G .*

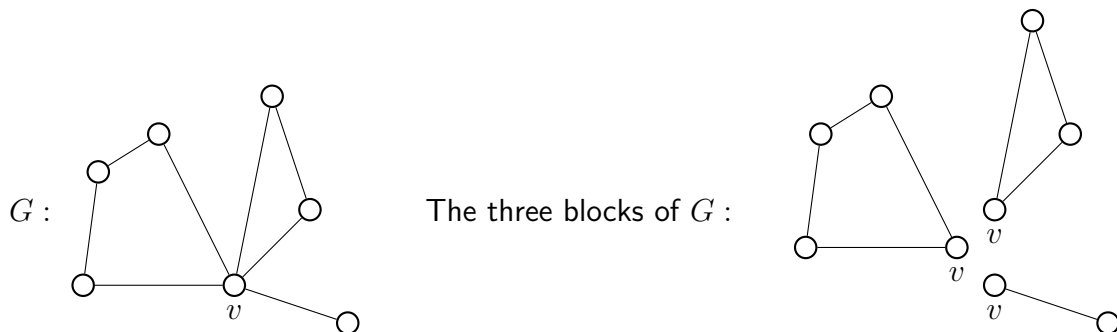
Proof. ($\implies$) Suppose first that $e = uv$ is a bridge of G . Then $G - e$ is disconnected, the vertices u and v lie in different components of $G - e$, and there is no $u - v$ path in $G - e$. Suppose now, to the contrary, that e lies on a cycle $C : u, v, v_1, v_2, \dots, v_k, u$ in G . But then $v, v_1, v_2, \dots, v_k, u$ is a $v - u$ path in $G - e$, a contradiction. Hence e does not lie on a cycle in G .

($\impliedby$) Suppose now that e does not lie on a cycle in G . Suppose, by way of contradiction, that e is not a bridge. Then $G - e$ is connected. Hence there is a $u - v$ path P in $G - e$. But then P together with the edge e is a cycle in G that contains the edge e , a contradiction. $\square$

Inspired by Theorem 11, we make the following definition: An edge e in a graph G is said to be a *cycle edge* if e lies on a cycle of G . Theorem 11 states that every edge of a connected graph is either a bridge or a cycle edge.

A connected graph that contains at least one cut-vertex is *separable*. A connected graph that contains no cut-vertices is called *non-separable*. A maximal non-separable subgraph of a connected graph is called a *block*, i.e., a block is a connected subgraph that has no cut-vertices and that is not properly contained in a subgraph with no cut-vertices. If G is non-separable, then G has no cut-vertices and hence G contains exactly one block. For this reason, a non-separable graph is sometimes called a *block*.

Example 3.3



The graph G from Example 3.2 has three blocks. The only cut-vertex of G , the vertex v , occurs in all three blocks. It's important to distinguish between the role the vertex v plays as a vertex of G and the role it plays as a vertex in each of the three blocks. In the graph G , the vertex v is a cut-vertex since $G - v$ is disconnected. On the other hand, let G' be any of the three blocks of G . Then $G' - v$ is *not* disconnected, so v is *not* a cut-vertex of G' . Notice, however, that in any connected subgraph of G that properly contains G' , the vertex v is a cut-vertex. Hence G' is a maximal connected subgraph that is non-separable, i.e., a block.

Two distinct blocks have at most one vertex in common. If two distinct blocks have a vertex in common, then that vertex is a cut-vertex. We now present some interesting characterizations of blocks. The proof of the first lemma is left as an exercise.

Lemma 12. *If G is a graph of order at least 3 that contains a bridge, then G contains a cut-vertex.*

Theorem 13. *A graph G of order at least 3 is a block if and only if for every pair u, v of vertices of G , there is a cycle containing both u and v .*

Proof. ($\Leftarrow$) Suppose that every pair of vertices of G lie on a common cycle and, to the contrary, that G contains a cut-vertex x . Then by Theorem 9, there are vertices u and v ($u, v \neq x$) such that x lies on every $u - v$ path. By assumption, the vertices u and v lie on a cycle C . However, since $C - x$ is connected, there is a $u - v$ path in G that does not contain the vertex x , a contradiction. Hence G contains no cut-vertex.

($\Rightarrow$) Suppose now that G has no cut-vertices and, to the contrary, that there is at least one pair of vertices that do not lie on a common cycle. Amongst all such pairs of vertices, choose u and v such that $d(u, v)$ is a minimum. By Lemma 12, no edge incident with u is a bridge. Hence, by Theorem 11, every edge incident with u lies on a cycle, i.e., every neighbour of u lies on a cycle containing u . Hence $d(u, v) \geq 2$. Let P be a $u - v$ geodesic and v' the vertex of P that is adjacent to v (since $d(u, v) \geq 2$, we are certain that $u \neq v'$). Since $d(u, v') < d(u, v)$, the vertex v' lies on a cycle C containing u . Furthermore, since v' is not a cut-vertex, there is a $v - u$ path P' that does not contain v' . Let t be the first vertex of P' that is on C . Let C' be the $t - v'$ path in C that contains u . By following P' from v to t , then C' from t through u to v' , then using the edge between v' and v , we obtain a cycle containing v . This is a contradiction. Hence no such pair u, v of vertices exists and the result is proved. $\square$

Let u and v be vertices in a connected graph G . Two $u - v$ paths P_1 and P_2 are called *internally disjoint* if $V(P_1) \cap V(P_2) = \{u, v\}$. Theorem 13 implies the following.

Corollary 14. *A graph G of order at least 3 is a block if and only if for every pair u, v of distinct vertices of G , there are (at least) two internally disjoint $u - v$ paths in G .*

The following result was proved by Frank Harary and Robert Norman in 1953 [HN53].

Theorem 15. *The center of a connected graph lies in a single block.*

Proof. Suppose, to the contrary, that there is a connected graph G and central vertices c_1, c_2 of G such that c_1 and c_2 do not lie in the same block. Let P be a $c_1 - c_2$ path. Then P is not contained in a single block. Consequently, P contains a vertex x which belongs to two different blocks, and hence x is a cut-vertex. Let G_1 and G_2 be the components of $G - x$ that contain, respectively, c_1 and c_2 . Let x' be an eccentric vertex¹ of x . Since x is a cut-vertex, there is at least one component, say G_1 , of $G - x$ that contains no vertices from any $x - x'$ path. This implies that $d(c_1, x') = d(c_1, x) + d(x, x') > d(x, x')$. This contradicts the fact that c_1 is a central vertex, and hence no such graph G exists. $\square$

¹If $d(x, x') = e(x)$, then x' is called an *eccentric vertex* of x .

Exercises

- 3.1 Prove that if G is a regular bipartite graph with partite sets V_1 and V_2 , then $|V_1| = |V_2|$.
- 3.2 Prove or disprove: For every positive integer t , the hypercube Q_t is bipartite.
- 3.3 Let G be a graph of order at least 5. Prove that at most one of G and $\overline{G}$ is bipartite.
- 3.4 Prove that a graph G is connected if and only if for every partition $\{V_1, V_2\}$ of $V(G)$, there is an edge of G joining a vertex of V_1 to a vertex of V_2 .
- 3.5 Prove that every nontrivial connected graph contains at least two vertices that are not cut-vertices.
- 3.6 Prove that if v is a cut-vertex of a connected graph G , then v is not a cut-vertex of $\overline{G}$.
- 3.7 Let G be a connected graph. Prove that a vertex v of G is a cut-vertex if and only if there are neighbours u and w of v such that v is on every $u - w$ path.
- 3.8 Let G be a connected graph in which every vertex has even degree. Prove that G does not contain a bridge.
- 3.9 Prove Lemma 12.
- 3.10 An *end-block* in a graph G is a block that contains exactly one cut-vertex of G . Prove that if G is separable, then G has at least two end-blocks.
- 3.11 Let G be a graph of order $n \geq 3$ in which, for every pair u, v of nonadjacent vertices, we have $\deg u + \deg v \geq n$. Prove that G is a block.
- 3.12
 - a. Prove or disprove: A graph G of order 3 or more is connected if and only if G contains two distinct vertices u and v such that $G - u$ and $G - v$ are connected.
 - b. Prove or disprove: Every connected graph of order 4 or more contains three distinct vertices u, v , and w such that $G - u, G - v$, and $G - w$ are connected.
- 3.13 What is the maximum number of cutvertices in a graph of order n ?
- 3.14 Let G be a cubic graph. Prove that G has a cutvertex if and only if G has a bridge.
- 3.15 Prove that every cubic graph with a bridge has order at least 10, and prove that this result is *sharp*, i.e., find a cubic graph of order 10 containing a bridge.
- 3.16 Prove that if G is a graph of order $n \geq 3$ with $\delta(G) \geq n/2$, then G is a block.
- 3.17 Prove or disprove: If G is a connected graph with cutvertices and u and v are vertices of G such that $d(u, v) = \text{diam } G$, then no block of G contains both u and v .
- 3.18 Let v be a vertex of a connected graph G . Prove that v is a cut-vertex of G if and only if there exists a partition $\{V_1, V_2\}$ of $V(G) - \{v\}$ such that for every pair v_1, v_2 of vertices with $v_1 \in V_1$ and $v_2 \in V_2$, every $v_1 - v_2$ path contains v .

- 3.19 Let G be a connected graph of order at least 3. Prove that G is a block if and only if for every vertex $v \in V(G)$ and edge $e \in E(G)$, there is a cycle containing both v and e .
- 3.20 If u is a cut-vertex in a graph G , prove that $G[N(u)]$ is disconnected. Is the converse true?
- 3.21 Prove that a graph is bipartite if and only if each of its components is bipartite.

Chapter 4

Trees

A graph that contains no cycles is *acyclic*. Graphs that are acyclic are also called *forests*. A *tree* is a connected acyclic graph. Hence, a forest is a graph in which each component is a tree. Figure 4 shows all of the trees of order at most 5.

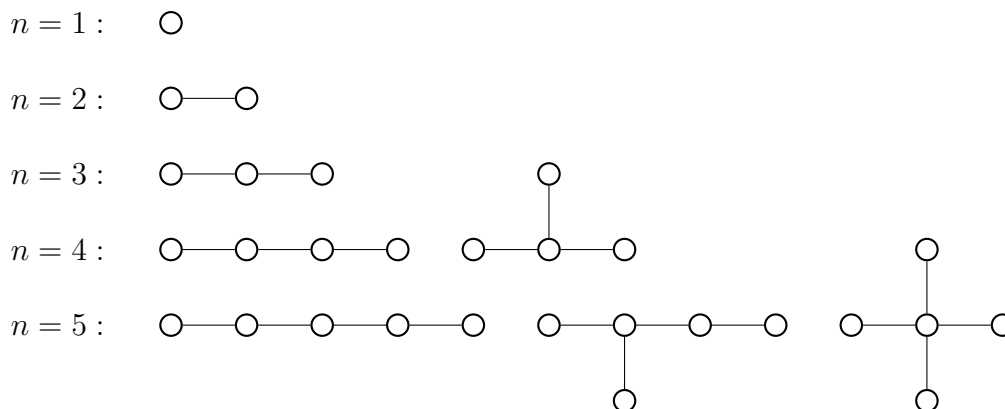


Figure 4.1: The trees of order $n \leq 5$.

Recall that a vertex of degree 1 is called an end-vertex.

Theorem 16. *Every nontrivial tree has at least two endvertices.*

Proof. Let T be a nontrivial tree and $P : v_1, v_2, \dots, v_k$ a maximal path in T (i.e., a path that is not properly contained in another path). If $\deg v_1 \geq 2$, then there is a vertex x which is adjacent to v_1 but, since T is acyclic, not on P . By starting at x , using the edge xv_1 to move to v_1 , and then the path P , we obtain a path that properly contains P , which is a contradiction. Hence, $\deg v_1 = 1$. Similarly, $\deg v_k = 1$. $\square$

Every block of order at least 3 contains a cycle. Since a tree is acyclic, every block in a nontrivial tree is isomorphic to K_2 . The next result therefore follows in a straightforward way from Theorem 15.

Theorem 17. *The center of a tree is isomorphic to K_1 or K_2 .*

A tree whose center is (isomorphic to) K_1 is sometimes called a *central tree*, while a tree whose center is (isomorphic to) K_2 is sometimes called a *bicentral tree*.

There are a number of well-known characterizations of trees.

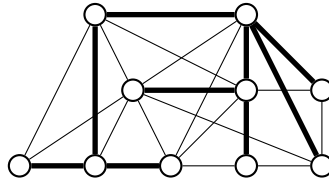
Theorem 18. *Every tree of order n has size $n - 1$.*

Proof. Our proof will be by induction on n . There is only one tree of order 1, and that tree (K_1) has size 0, so when $n = 1$ the statement is true. Suppose now that $n > 1$ and that the statement is true for every tree of order less than n . Let T be a tree of order n and size m . By Theorem 16, T has an end-vertex v . No end-vertex is a cut-vertex, so $T - v$ is a tree of order $n(T - v) = n - 1$ and size $m(T - v) = m - 1$. By the Inductive Hypothesis, $m(T - v) = n(T - v) - 1$. Hence $m - 1 = n - 1 - 1$, which implies that $m = n - 1$. $\square$

Corollary 19. *A forest of order n with k components has size $n - k$.*

If G is a connected graph, then a *spanning tree* of G is a spanning subgraph of G that is a tree.

Example 4.1



In the connected graph shown above, the edges of a spanning tree are indicated as darker lines.

Theorem 20. *Every connected graph has a spanning tree.*

Theorem 21. *Let G be a graph of order n and size m . If any two of the following three statements are true, then G is a tree:*

- (i). G is connected.
- (ii). G is acyclic.
- (iii). $m = n - 1$.

Proof. ((i) and (ii)) If G is connected and acyclic, then by definition G is a tree.

((i) and (iii)) Suppose that G is connected and that $m = n - 1$. By Theorem 20, the graph G has a spanning tree T . By Theorem 18, the tree T has size $n - 1$, the same as G . Since T is a subgraph of G , we must therefore have $G = T$, i.e., G is a tree.

((ii) and (iii)) Since G is acyclic, G is a forest. Let k be the number of components of G . Then, by Corollary 19, $k = 1$, i.e., G is connected. Hence, G is a tree. $\square$

Theorem 22. *If T is a tree of order k and G is a graph with $\delta(G) \geq k - 1$, then T is a subgraph of G .*

Proof. Our proof will be by induction on k . Certainly the result is true when $k = 1$. Suppose now that $k > 1$ and that the result is true for trees of order less than k , i.e., if T is a tree with $n(T) < k$ and G is a graph with $\delta(G) \geq n(T) - 1$, then T is a subgraph of G . Let T be a tree of order k and G a graph with $\delta(G) \geq k - 1$. Let v be an end-vertex of T and v' the neighbour of v . Then $T - v$ is a tree of order less than k . Since $\delta(G) \geq k - 1 > n(T - v) - 1$,

by the Inductive Hypothesis, the tree $T - v$ is a subgraph of G . In the graph G , the vertex v' has degree at least $\delta(G) \geq k - 1$. However, in the tree $T - v$, the vertex v' has degree at most $k - 2$. Hence there is a vertex x of G to which v' is adjacent in G but not adjacent in $T - v$. Then by adding the vertex x and the edge xv' to $T - v$ we obtain a tree that is isomorphic to T and contained in G . $\square$

Exercises

- 4.1 Draw all trees of order 6.
- 4.2 Prove that if every vertex of a graph G has degree at least 2, then G contains a cycle.
- 4.3 Prove or disprove: If G is a graph of order n and size m with three cycles, then $m \geq n + 2$.
- 4.4 Find a graph of order n and size $n - 1$ that is not a tree.
- 4.5 Prove that if G is a connected graph of order n and size m , then $m \geq n - 1$.
- 4.6 Suppose that T is a tree of order $n \geq 4$ that contains only vertices of degree 1 and 3. Prove that T contains $(n - 2)/2$ vertices of degree 3.
- 4.7 Prove Corollary 19.
- 4.8 Prove Theorem 20.
- 4.9 Prove that the following statements are equivalent for an (n, m) graph G :
- G is a tree.
 - Every two vertices of G are joined by a unique path.
 - G is connected and $m = n - 1$.
 - G is acyclic and $m = n - 1$.
 - G is acyclic and for every pair u, v of distinct nonadjacent vertices of G , the graph $G + uv$ has exactly one cycle.
- 4.10 A graph that is connected and has exactly one cycle is called *unicyclic*. Prove that the following four statements are equivalent:
- G is unicyclic.
 - G is connected and $m = n$.
 - For some edge e of G , the graph $G - e$ is a tree.
 - G is connected and the set of edges that are not bridges form a cycle.
- 4.11 Prove that a tree of order at least 3 has diameter 2 if and only if it is a star.
- 4.12 Prove or disprove:
- If G has diameter 2, then it has a spanning star.
 - If G has a spanning star, then it has diameter 2.
- 4.13 Prove that G is a forest if and only if every induced subgraph of G contains a vertex of degree at most 1.
- 4.14 Prove Theorem 17.

Chapter 5

Planar graphs

5.1 Jordan's Theorem and Euler's Formula

If a graph G can be drawn on a sheet of paper¹ in such a way that none of the edges cross, then G is a *planar* graph, otherwise it is *nonplanar*. Drawing a graph on a sheet of paper in such a way that none of the edges cross is equivalent to finding an *embedding* of the graph in the plane, that is, a map that associates the vertices and edges of G with distinct points and curves² in the plane in such a way that two points in the plane are the endpoints of a curve if and only if they correspond to two adjacent vertices. In fact, the following is true (the proof is left as a homework problem).

Theorem 23. *A graph can be embedded in the plane if and only if it can be embedded on the sphere.*

An embedding of a graph in the plane or on the sphere is called a *plane graph*. Every plane graph can be considered a planar graph, but not every drawing of a planar graph is plane. It is clear that a graph is planar if and only if each of its components is planar, so we shall in general restrict our attention to connected graphs.

Example 5.1



The graph shown here, K_4 , is drawn on the left with crossing edges. On the right, we have redrawn it so that none of the edges cross. Since K_4 can be embedded in the plane, as shown on the right, it is planar. However, only the right-hand drawing of K_4 is a plane graph.

¹It's necessary to qualify where G is to be drawn because graph theorists also draw graphs on other things, like spheres and donuts and, more generally, orientable and non-orientable surfaces of different genus.

²A *curve* in the plane is a continuous function $f : [0, 1] \rightarrow \mathbb{R}^2$. A curve is *closed* if $f(0) = f(1)$ and *simple* if f is injective on $[0, 1)$.

We begin our study of planar graphs with a famous theorem that we shall not prove.

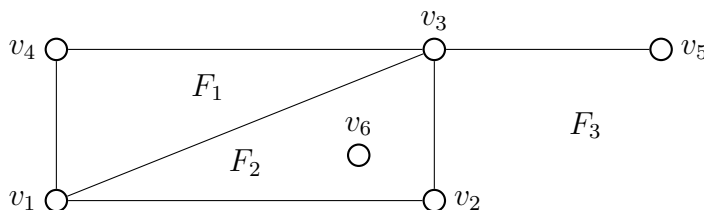
Theorem 24 (Jordan Curve Theorem). *A simple closed curve C divides the plane into two regions, exactly one of which is bounded, with C the boundary of both.*

Jordan's Theorem was first proved by Camille Jordan in the 3rd edition of his text on analysis [Jor15]. A modern proof of Jordan's Theorem can be found in a paper by Thomassen [Tho92], who remarks:

‘Although this fundamental result seems intuitively obvious it is fascinatingly difficult to prove’.

Consider a plane graph, G . By Jordan's Theorem, every cycle C in G divides the plane into three disjoint sets: The cycle C , the *interior* of C , and the (unbounded) *exterior* of C . The maximal connected portions of the plane that remain when we delete the (points and curves corresponding to the) vertices and edges of G are the *faces* or *regions* of G . Exactly one face of G is unbounded: this is the *outer face* or *exterior face*. Every other face is an *inner face* of G . Let $S \subseteq \mathbb{R}^2$. A point $x \in \mathbb{R}^2$ is called a *boundary point* of S if every open set containing x has a nonempty intersection with both S and $\mathbb{R}^2 - S$. The *boundary* of S is the set of all its boundary points. If F is a face of a plane graph G , then the *boundary* of F is the subgraph $G[F]$ of G consisting of those vertices and edges that correspond to boundary points of the set F .

Example 5.2



The plane graph shown here has three faces: F_1 , F_2 , and F_3 . The boundary $G[F_1]$ of F_1 is the cycle v_1, v_3, v_4, v_1 (including its edges). Similarly, $G[F_2]$ is the cycle v_1, v_2, v_3, v_1 together with the vertex v_6 , and the boundary of the outer face F_3 is the cycle v_1, v_2, v_3, v_4, v_1 together with the vertex v_5 and the edge v_3v_5 .

Our next result is commonly referred to as Euler's Formula. It was noted (but not rigorously proved) by Leonhard Euler in 1751, but had previously been mentioned by René Descartes in 1639. Many proofs of Euler's Formula have been found³. We shall give only one.

Theorem 25 (Euler's Formula). *If G is a connected plane graph with n vertices, m edges, and f faces, then $n - m + f = 2$.*

Proof. The proof will be by induction on m . When $m = 0$, then $G \cong K_1$. Hence $n = 1$, there is one face, and $n - m + f = 2$. Suppose now that the result is true for all connected plane graphs of size less than m and let G be a connected plane graph of size m . If G is a tree, then by Theorem 18, $m = n - 1$ and, since G is acyclic, there is exactly one face. Hence $n - m + f = n - (n - 1) + 1 = 2$, as required. If G is not a tree, then G contains a cycle. Let e

³Have a look at *Twenty Proofs of Euler's Theorem* at www.ics.uci.edu/~eppstein/junkyard/euler/.

be a cycle edge of G . Then by Jordan's Theorem e is on the boundary of two faces in G , and in $G - e$ these two faces become one face, so the graph $G - e$ has $f - 1$ faces. By Theorem 11, the edge e is not a bridge, so $G - e$ is a connected plane graph of size $m - 1$ (and order n). By the Inductive Hypothesis, we then have $n - (m - 1) + (f - 1) = 2$, and hence $n - m + f = 2$. $\square$

Euler's Formula implies that if G is a planar graph, then, however we embed G in the plane, the number of faces does not change. Hence we can speak of the number of faces of a planar graph. Things get a little more complicated, though, if we ask whether we can speak of 'the faces' of a planar graph. In general, a planar graph may admit embeddings that are not 'isomorphic' (Example 5.3). However, Whitney [Whi32] proved that every 3-connected⁴ planar graph has essentially only one embedding. For a rigorous discussion of isomorphisms between plane graphs and a proof of Whitney's Theorem, see [Die05].

Example 5.3



Two 'nonisomorphic' embeddings of the same planar graph G : The plane graph G'' has a face with boundary C_5 ; the plane graph G' has no such face.

5.2 Maximal planar graphs

If G is a planar graph but the addition of any edge to G produces a graph that is nonplanar, then G is a *maximal planar graph*. A plane embedding of a maximal planar graph is a *maximal plane graph*. The proof of the following lemma is left as an exercise.

Lemma 26. *Let G be a maximal plane graph of order at least 3 and F a face of G .*

- (a) $G[F] = K_3$ (i.e., the boundary of every face of G is a triangle).
- (b) If $xy, xz \in E(G[F])$, then $yz \in E(G[F])$.

We now use Lemma 26 to establish an important property of maximal planar graphs.

Theorem 27. *If G is a maximal planar graph of order $n \geq 3$ and size m , then $m = 3n - 6$.*

Proof. Consider an embedding of G in the plane and denote by f the number of faces of G . For a face F , let $\epsilon(F)$ be the number of edges in $G[F]$. By Lemma 26, each face of G is a triangle, and hence $\sum_F \epsilon(F) = 3f$. Moreover, each edge is on the boundary of exactly two faces, so $\sum_F \epsilon(F) = 2m$. Consequently, $f = \frac{2}{3}m$. Substituting this into Euler's Formula and rearranging the terms, we obtain the desired result. $\square$

⁴A connected graph G is k -connected if for every $(k - 1)$ -subset S of $V(G)$, $G - S$ is connected. So a 2-connected graph is a connected graph with no cut-vertices, and a 3-connected graph is still connected after the removal of any pair of vertices.

If G is a planar graph of order n , then clearly G is a subgraph of a maximal planar graph of order n . Hence, we obtain the following corollary.

Corollary 28. *If G is a planar graph of order $n \geq 3$ and size m , then $m \leq 3n - 6$.*

We shall see later that the graphs K_5 and $K_{3,3}$ play an important role in the theory of planarity. We can use Corollary 28 to prove that K_5 is nonplanar.

Example 5.4 The complete graph K_5 has order $n = 5$ and size $m = 10 > 9 = 3n - 6$. So, by Corollary 28, K_5 is nonplanar.

What about $K_{3,3}$? Here, $n = 6$ and $m = 9 < 12 = 3n - 6$, so Corollary 28 tells us nothing. However, by considering the structure of $K_{3,3}$ in slightly more detail and modifying the proof of Theorem 27, we can prove that $K_{3,3}$ is also nonplanar.

Example 5.5 Suppose, to the contrary, that $K_{3,3}$ is planar and consider an embedding of $K_{3,3}$ in the plane. Since $K_{3,3}$ is bipartite, by Theorem 8 it contains no odd cycle. No edge of $K_{3,3}$ is a bridge, so by Theorem 11 every edge of $K_{3,3}$ lies on a cycle. Hence every face of $K_{3,3}$ is bounded by a cycle of length at least 4, and every edge is on the boundary of two faces. Thus, $4f \leq \sum_F \epsilon(F) = 2m$, which implies, on substitution into Euler's Formula, that $m \leq 2n - 4$. However, for $K_{3,3}$, we have $m = 9 > 8 = 2n - 4$. This is a contradiction, so $K_{3,3}$ is nonplanar.

An additional useful consequence of Corollary 28 is the following result, originally due to Kempe [Kem79] in his ill-fated attempt to prove the Four Colour Theorem.

Lemma 29. *If G is a planar graph, then $\delta(G) \leq 5$.*

Proof. Suppose that G has order n , size m , and, to the contrary, minimum degree $\delta(G) \geq 6$. Then, by Theorem 1, we have

$$2m = \sum_{v \in V(G)} \deg v \geq 6n.$$

Hence $m \geq 3n$, contradicting Corollary 28. □

Theorem 30. *Let u be a vertex in a maximal planar graph G of order at least 4. Then there is a unique cycle C of G with $V(C) = N(u)$.*

Proof. First we prove that there is at least one cycle through the vertices of $N(u)$. Consider an embedding of G in the plane, and suppose that the edges incident with u are $uv_1, uv_2, \dots, uv_k$ and that they occur cyclically in that order as we proceed clockwise around u . Then for all $i \in [1, k]$, the edges uv_i and uv_{i+1} (the subscripts read modulo k) lie on the boundary of a common face, so by Lemma 26 we have $v_i v_{i+1} \in E(G)$. Hence there is at least one cycle $v_1, v_2, \dots, v_k, v_1$ through $N(u)$.

Now suppose, to the contrary, that C and C' are cycles with $V(C) = N(u) = V(C')$ and that $C \neq C'$. Consider an embedding of G in the plane in which u lies in the interior of C . Then there are vertices $x, y \in N(u)$ with $xy \in E(C') - E(C)$. Since $xy \notin E(C)$, there are neighbours z_1 and z_2 of u that lie in the interior and exterior, respectively, of the triangle uxy and that are adjacent to x . There are two internally disjoint $z_1 - z_2$ paths that collectively use all the edges of C' ; necessarily, one includes the vertex x while the other includes the vertex y . Since these paths are internally disjoint, however, neither may use the edge xy of C' , a contradiction. □

Following on from Theorem 30, we'll henceforth use $C(u)$ for the unique cycle through the neighbours of a vertex u in a maximal planar graph, and refer to this as the u -cycle.

Theorem 31. *If G is a maximal planar graph of order at least 4, then $3 \leq \delta(G) \leq 5$.*

Proof. The upper bound is from Lemma 29. For the other part: Suppose, to the contrary, that there is a maximal planar graph G of order $n \geq 4$ and size m with a vertex v of degree at most 2. Let G' be the graph $G - v$ of order $n' = n - 1$ and size $m' \geq m - 2$. By Theorem 27, $m = 3n - 6$, and hence $m' \geq m - 2 = 3n - 8 = 3(n' + 1) - 8 = 3n' - 5$, contradicting the fact that G' is planar. $\square$

The proof of the following lemma is an exercise earlier in the notes.

Lemma 32. *If u is a cut-vertex in a graph G , then $G[N(u)]$ is disconnected.*

5.3 Chvátal's Art Gallery Theorem and Fáry's Theorem

If you draw some small plane graphs, you will see that you can always find a drawing in which each edge is a straight line segment. Such a drawing is called a *straight line embedding*. In fact, every planar graph has a straight line embedding. This result, commonly known as Fáry's Theorem, was proved independently by Klaus Wagner [Wag36] in 1936, István Fáry [Fár48] in 1948, and Sherman Stein [Ste51] in 1951. The proof that we shall give employs a well-known result due to Vašek Chvátal [Chv75] named *Chvátal's Art Gallery Theorem* or *Watchman Theorem*. A few years after the Art Gallery Theorem first appeared in the literature, Steve Fisk found a proof [Fis78] so elegant that it's included in Martin Aigner and Günter Ziegler's *Proofs from The Book* [AZ18]. To give Fisk's proof, we must first define something we'll consider in more depth later on: If G is a graph, then a *colouring* of G is an assignment of colours to the vertices of G , i.e., a function $c : V(G) \rightarrow S$ for some set S of colours. If $|c(V(G))| = k$, then c is a k -colouring. If c is a colouring of G with the property that $uv \in E(G)$ implies that $c(u) \neq c(v)$, then c is a *proper colouring*⁵ of G .

Theorem 33 (Chvátal's Art Gallery Theorem). *If P is a simple polygon with n vertices, then there is a set S of at most $\lfloor n/3 \rfloor$ vertices of P such that every point of P is visible from S .*

Proof. [Fis78] If $n = 3$ the result is obviously true, so we assume that $n \geq 4$. Let G be a graph obtained from P by triangulating its interior and xy an edge of G that does not lie on the boundary of the exterior face. Colour x and y with the colours 1 and 2, then repeatedly colour each triangle in G that has two colours by colouring the remaining vertex with the missing colour. This produces a proper 3-colouring of G . One colour class — say the set of red vertices — has cardinality at most $\lfloor n/3 \rfloor$ and every triangle contains a red vertex. Since triangles are convex, every point of G is visible from the set of red vertices. $\square$

The proof of the next result is left as a homework exercise.

Corollary 34. *If P is a simple polygon with n vertices, then there is a set S of at most $\lfloor n/3 \rfloor$ points in the interior of P such that every point of P is visible from S .*

⁵Proper colourings are one of many different kinds of colouring that have been studied. There exist books that deal solely with different colourings of graphs (see, for example, [JT95] or [CZ09]).

Theorem 35 (Fáry’s Theorem). *Every planar graph has a straight line embedding.*

Proof. By induction on the order. If G is a planar graph of order 3 or less, then G is a subgraph of K_3 , which is the unique maximal planar graph of order 3 and which clearly has a straight line embedding, so G has a straight line embedding. Assume then that every planar graph of order n or less, where $n \geq 3$, has a straight line embedding. Let G be a planar graph of order $n + 1$, and let G' be a maximal planar graph of order $n + 1$ that contains G . By Theorem 31, G' contains a vertex v with $\deg v \in \{3, 4, 5\}$. The graph $G'' = G' - v$ is a planar graph of order n and hence has a straight line embedding in the plane, which we choose in such a way that the face F bounded by $C(v)$ is not the exterior face. Then $G''[F]$ is a simple polygon with at most 5 vertices, so by Corollary 34 there is a point $x \in F$ from which all of the vertices of $C(v)$ are visible. We then put v at x and draw straight lines to the vertices of $C(v)$ to get a straight line embedding of G' and hence of G . $\square$

5.4 The Four Colour Theorem

The *chromatic number* of a graph G is the minimum number of colours in a proper colouring of G , i.e., the minimum k for which G has a proper k -colouring. We denote the chromatic number of G by $\chi(G)$. A graph with chromatic number k is called *k -chromatic*. If G has a proper colouring using at most k colours, then G is called *k -colourable*. Clearly, G is 2-colourable if and only if G is bipartite.

Example 5.6



Above are shown two proper colourings of the same graph G using the colours white, black, light grey, and dark grey: On the left, a proper 4-colouring; on the right, a proper 3-colouring. Hence G is 3-colourable and 4-colourable. By Theorem 8, the graph G is not bipartite and there is no proper 2-colouring of G . From this we deduce that $\chi(G) = 3$.

⁶On 23 October 1852, an English mathematics student named Frederick Guthrie⁷ asked his professor, the famous mathematician Augustus de Morgan, whether it is true that every planar graph is 4-colourable⁸. De Morgan found Guthrie’s Four Colour Problem interesting and later that day wrote to the famous mathematician Sir William Rowan Hamilton to inquire as to

⁶The description that follows is based largely on that given in the two books [Wil02] and [FF98].

⁷Frederick Guthrie got the problem from his brother Francis Guthrie, who was then a law student in London. Francis Guthrie later became Professor of Mathematics at the University of Cape Town, then known as the South African College. He also lectured on botany, inspiring Harry Bolus — who later founded UCT’s *Bolus Herbarium* — to take up the subject. He died shortly after retiring in 1899 and is buried in the cemetery of St Thomas Anglican Church in Campground Road, Rondebosch. The story of Guthrie’s life is told in detail in [MM12].

⁸The question he asked was actually whether *the greatest necessary number of colours to be used in colouring a map so as to avoid identity of colour in lineally contiguous districts is four* [Gut80]. This is equivalent to the question of whether planar graphs are 4-colourable.

whether he knew a proof. Hamilton was uninterested, but the problem gained some attention when, in 1878, the famous mathematician Arthur Cayley asked in a meeting of the London Mathematical Society⁹ whether a solution to the Four Colour Problem had been found. One year later, a lawyer named Alfred Bray Kempe published a proof [Kem79] of the Four Colour Theorem, but in 1890 a mathematician named Percy John Heawood announced that he had found a mistake in Kempe's proof. Despite a great deal of effort over the years that followed, the Four Colour Problem remained unsolved until 1976, when two mathematicians at the University of Illinois at Urbana-Champaign, Kenneth Appel and Wolfgang Haken, announced that they had found a proof. The proof involved the checking of an extremely large number of cases, however, for which they used 1200 hours of time on an IBM 360 mainframe computer. A number of small errors in the proof were found, and corrected, over the years that followed. In 1989, a corrected version of the proof was published as a 741 page book [AH89] by the American Mathematical Society.

Theorem 36 (The Four Colour Theorem). *Every planar graph is 4-colourable.*

Although Kempe's proof was incorrect, some of the ideas in his proof¹⁰ can be used to prove a much easier result: That every planar graph is 5-colourable.

Theorem 37 (The Five Colour Theorem). *Every planar graph is 5-colourable.*

Proof. Our proof will be by induction on the order, n , of a planar graph G . If G has order at most 5, then the result follows trivially. For some integer $n \geq 6$, assume that the result holds for all planar graphs of order at most $n - 1$, i.e., that every planar graph of order at most $n - 1$ is 5-colourable. Let G be a planar graph of order n . By Lemma 29, G has a vertex v of degree at most 5. Since $G - v$ is a planar graph of order $n - 1$, by the Inductive Hypothesis it has a proper colouring using at most 5 colours. If one of the five colours is not used in the neighbourhood of v , then v may be coloured with that missing colour, producing a proper 5-colouring of G . We assume, therefore, that $\deg v = 5$ and that all five colours, which we shall call 1,2,3,4,5, are used in the neighbourhood of v . Consider an embedding of G in the plane and assume, without loss of generality, that the neighbours of v , which we denote $v_1, v_2, \dots, v_5$, are arranged cyclically around v in the plane (see Figure 5.1) so that each v_i is coloured with colour i and the colours are used in the order 1,2,3,4,5. Let $G(1,3)$ be the subgraph of G

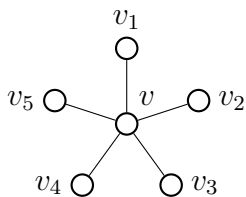


Figure 5.1: The neighbourhood of the vertex v in the proof of Theorem 37.

induced by all the vertices that are coloured 1 or 3. If v_1 and v_3 are in different components of $G(1,3)$, then by swapping the colours of the vertices in the component of $G(1,3)$ containing

⁹The founding president of which was De Morgan.

¹⁰Kempe was the first to see that every planar graph has a vertex of degree 5 or less. He also devised the *Kempe Chains* that play a role in the proof of the Five Colour Theorem.

v_1 (i.e., all the vertices that were coloured 1 are now coloured 3, and vice-versa) we produce a colouring in which the colour 1 is not used at any neighbour of v , and so v may be coloured with the colour 1 to produce a proper 5-colouring of G . Suppose, then, that v_1 and v_3 are in the same component of $G(1, 3)$. Then there is a path P from v_1 to v_3 consisting only of vertices coloured with the colour 1 or the colour 3. The path P together with v and the edges vv_1 and vv_3 is then a cycle bounding a region containing the vertex v_2 . It follows that there is no path from v_2 to v_4 consisting only of vertices coloured 2 or 4. Let $G(2, 4)$ be the subgraph of G induced by the vertices that are coloured 2 or 4. By swapping the colours of the vertices in the component of $G(2, 4)$ that contains v_2 , we obtain a proper 5-colouring of $G - v$ in which the colour 2 is not used at any neighbour of v . By colouring v with the colour 2, we then obtain a proper 5-colouring of G . $\square$

The proof of the Four Colour Theorem is a proof by *minimum counterexample*, which is a mixture of proof by induction and proof by contradiction. Assume, to the contrary, that the Four Colour Theorem is not true. Then there is at least one planar graph that is not 4-colourable. Amongst all such planar graphs, let G be a *smallest* counterexample, i.e., G is a planar graph with $\chi(G) = 5$ and every proper subgraph of G is 4-colourable. We now argue:

1. That G must contain a subgraph that is in an *unavoidable set* of 1,936 ‘configurations’¹¹.
2. That for each of these 1,936 configurations, we can either replace it with something smaller (violating the minimality of G) or find a way of extending the proper 4-colouring of a proper subgraph of G to a proper 4-colouring of the whole of G (in the language of the proof, we are proving that each configuration is *reducible*), contradicting our choice of G .

What do we mean by an *unavoidable set of configurations*? By Lemma 29, we may assume that G has a vertex v with $\deg v \leq 5$. In fact, if $\deg v \leq 4$, then the proof of the Five Colour Theorem shows that G is 4-colourable. Thus, if G is a counterexample to the Four Colour Theorem, we can assume that G contains a vertex of degree 5: i.e., ‘ G has a vertex of degree 5’ is *unavoidable*. We represent this with the statement that $\{\bullet\}$ is an unavoidable set of configurations, where $\bullet$ is shorthand for ‘ G contains a vertex of degree 5’. In Kempe’s incorrect proof, he attempts to show that this configuration is reducible by a similar argument to the proof of the Five Colour Theorem, i.e., by arguing that a proper 4-colouring of $G - v$ can be extended to a proper 4-colouring of G . Heawood, however, found (in modern terminology) a graph on which Kempe’s argument did not work as intended [Hea90]. Heawood was able to use the ideas in Kempe’s proof to prove the Five Colour Theorem. Despite a lot of effort by Heawood and others, no-one could find a way to fix the mistake in Kempe’s proof, i.e., to prove that the unavoidable set $\{\bullet\}$ is also reducible.

Over the years, that followed, progressively more sophisticated arguments were developed in an attempt to find a set of configurations that is both unavoidable and reducible. The vertices of degree 5 play a crucial role that we now describe. If v is a vertex in a graph G , then the *charge* of v is

$$q(v) = 6 - \deg v.$$

Thus, vertices of degree 5 have charge +1, those of degree 6 have charge 0, and those of larger degree have negative charge. If G is a planar graph of order n and size m , we have

¹¹In 1997, Neil Robertson, Daniel Sanders, Paul Seymour, and Robin Thomas [RSST97] managed to simplify Appel and Haken’s proof, producing a set of 633 configurations that are both unavoidable and reducible.

$6n - 12 \geq 2m = \sum_{v \in V(G)} \deg v$, and hence

$$\sum_{v \in V(G)} q(v) \geq 12.$$

Suppose that G is a minimum counterexample to the Four Colour Problem. As discussed, we may assume that G has no vertex of degree less than 5. Since the total charge on G is positive, there are at least twelve vertices of degree 5. We can exploit this, in various ways, to construct unavoidable sets of configurations. We give one example below. We use $\bigcirc$ to indicate a vertex of degree 6.

Theorem 38. *Let G be a maximal planar graph with $\delta(G) = 5$. Then $\{ \bullet \text{---} \bullet , \bullet \text{---} \bigcirc \}$ is an unavoidable set of configurations.*

Proof. Suppose to the contrary that there exists a maximal planar graph G with $\delta(G) = 5$ and where no vertex of degree 5 is adjacent to a vertex of degree 5 or 6. Consider the following *discharging operation*: For each vertex u of degree 5, discharge $1/5$ from u to each neighbour with a degree of 7 or more. Since no vertex of degree 5 is adjacent to a vertex of degree 5 or 6, each such vertex is fully discharged. If v is a vertex of degree $k \geq 7$, then no more than $k/2$ of the neighbours of v have degree 5, so after moving the charge the total charge on v is at most $6 - k + k/2 \cdot 1/5 = 6 - 9k/10$, which is negative since $k \geq 7$. But now every vertex of G has nonpositive charge, implying that the total charge on G is nonpositive, a contradiction. $\square$

If you're interested in reading more about discharging and the proof of the Four Colour Theorem, I recommend the book by Wilson [Wil02].

Exercises

- 5.1 Give an example of a planar graph in which no vertex has degree less than 5.
- 5.2 If G is a planar graph of order at least 4, prove that G contains at least four vertices of degree at most 5.
- 5.3 Prove that there exists only one 4-regular maximal planar graph.
- 5.4 Let $k \geq 3$ be an integer and let G be a plane graph of order $n \geq k$ and size m in which each cycle has length at least k . Determine an upper bound for m in terms of n and k .
- 5.5 Prove Theorem 23.
- 5.6 If G is a plane graph and F is any face of G , prove that G can be embedded in the plane in such a way that F is the outer face.
- 5.7 Euler's Formula applies only to connected planar graphs. Generalize Euler's Formula so that it can be used with any planar graph.
- 5.8 A graph G is called *outerplanar* if there is an embedding of G in the plane such that every vertex of G lies on the boundary of the outer face.
- Prove that G is outerplanar if and only if $G + K_1$ is planar.
 - If G is an outerplanar graph of order $n \geq 2$ and size m , prove that $m \leq 2n - 3$.
- 5.9 For each $k \in \{1, 2, 3\}$, find a nonplanar graph with chromatic number k , or prove that no such graph exists.
- 5.10 Recall that the average distance of a connected graph G is $\mu(G) = \frac{2}{n(n-1)} \sum_{u,v \in V(G)} d(u, v)$.
If G is a maximal planar graph of order n and diameter 2, find a formula for $\mu(G)$.
- 5.11 Prove Lemma 26.
- 5.12 Prove that the average degree of the vertices of a planar graph is less than 6.
- 5.13 If G is a maximal planar graph with $\delta(G) = 5$, use the discharging operation 'Every vertex of degree 5 gives a charge of $1/3$ to each of its neighbours of degree 7 or more' to prove that
-  is an unavoidable set of configurations.
- 5.14 If G is a maximal planar graph with $\delta(G) = 3$, prove that G contains K_4 as a subgraph.
- 5.15 Determine for which values of t the hypercube Q_t is planar.

Part II

Not Graph Theory

Appendix A

Permutations, Combinations, and the Binomial Theorem

NOTE: Everything in this chapter was covered in MAM1000W / MAM1032S/ MAM1034S!

A.1 Factorials

Definition A.1 For a nonnegative integer n , we define $n!$ (pronounced n factorial) in the following way: $0! = 1$, and for $n \geq 1$,

$$n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1.$$

Example A.2

$$0! = 1$$

$$1! = 1$$

$$2! = 2 \cdot 1 = 2$$

$$3! = 3 \cdot 2 \cdot 1 = 6$$

$$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$$

It's useful to remember that

$$n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1 = n \cdot [(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1] = n \cdot (n-1)!$$

Example A.3

$$5! = 5 \cdot 4! = 5 \cdot 24 = 120.$$

We'll use the fact that $n! = n \cdot (n-1)!$ in the next example.

Example A.4 Use mathematical induction to prove that for every positive integer n ,

$$\frac{d^n}{dx^n}(x^n) = n!.$$

If $n = 1$, then

$$\frac{d^1}{dx^1}(x^1) = \frac{d}{dx}(x) = 1 = 1!,$$

which proves the Base Case. Suppose now that the statement is true for some integer $n \geq 1$, i.e.,

$$\frac{d^n}{dx^n}(x^n) = n!$$

Then

$$\begin{aligned} \frac{d^{n+1}}{dx^{n+1}}(x^{n+1}) &= \frac{d^n}{dx^n} \left(\frac{d}{dx}(x^{n+1}) \right) \\ &= \frac{d^n}{dx^n} ((n+1)x^n) \\ &= (n+1) \frac{d^n}{dx^n}(x^n) \\ &= (n+1) \cdot n! \quad (\text{by the Inductive Hypothesis}) \\ &= (n+1)! \end{aligned}$$

We have proved that the statement is true when $n = 1$, and we have proved that whenever the statement is true for some integer $n \geq 1$, it is also true for the integer $n+1$. Hence, by the Principle of Mathematical Induction, the statement is true for every positive integer n .

Factorials are important in counting, as the next example shows.

Example A.5 In how many different ways can we arrange the letters a , b , and c ?

Here are all the possibilities:

abc
 acb
 bac
 bca
 cab
 cba

So the answer is 6.

Here's a different way of thinking about this problem: When we're arranging the letters, we first pick which letter will go first, then we pick which letter will go second, and then whichever letter is left over must go third. There are three options — a , b , or c — for which letter will go first. Whichever we pick, there are then two options for which letter goes second (e.g., if we choose b to

go first, then we can pick a or c to go second). So the total number of possible arrangements of these three letters is

$$3 \cdot 2 \cdot 1 = 3!$$

The preceding example illustrates the following important result:

Theorem 39. *The number of different ways of arranging n objects is $n!$.*

Example A.6 *In how many different ways can six male students queue to go to the bathroom before a MAM1000W test?*

Whichever of the six students is first in the queue, one of the remaining five must be second, then one of the other four must be third, then one of the last three must be fourth, one of the final two must be fifth, and then the unluckiest student will go to the end of the queue. So the number of possible queues is

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720.$$

You might have noticed that $n!$ grows quickly, but how quickly exactly? A commonly used approximation for $n!$ is *Stirling's Approximation*:

$$n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n.$$

This shows that $n!$ grows exponentially fast with n . The derivation of Stirling's Approximation is usually covered in second or third year.

A.2 One-to-one functions and permutations

Throughout this section, we shall denote the domain and range of a function f as $D(f)$ and $R(f)$, respectively. As you already know, a function is *one-to-one* if no two x -values have the same y -value (or, equivalently, if its graph passes the Horizontal Line Test). More formally, a function f is one-to-one if for all $x_1, x_2 \in D(f)$, if $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$.

Example A.7 As you no doubt recall, the function $f : \mathbb{R} \rightarrow \mathbb{R}$ with rule $f(x) = x^2$ is not one-to-one. It's not one-to-one because we can find two numbers $x_1, x_2 \in \mathbb{R}$ where $x_1 \neq x_2$ but $f(x_1) = f(x_2)$. For example,

$$f(-1) = (-1)^2 = 1 = 1^2 = f(1).$$

Definition A.8 A *permutation* of a set S is a one-to-one function f with $D(f) = S = R(f)$.

Example A.9 Let f be the function with domain $[0, 1]$ and rule $f(x) = 1 + x$. Then f is one-to-one (why?), but f is not a permutation because $D(f) = [0, 1]$ but $R(f) = [1, 2]$.

Example A.10 Let g be the function with domain $[0, 1]$ and rule $g(x) = 1 - x$. Then g is one-to-one (why?), and $D(g) = [0, 1] = R(g)$, so g is a permutation of $[0, 1]$.

In this section, we shall mostly be concerned with permutations of finite sets (sets that have a finite number of elements). An important point about these kinds of permutations is that we can use a convenient shorthand to write them down, as we see in the next example.

Example A.11 Let $f : \{1, 2, 3\} \rightarrow \{1, 2, 3\}$ be the function with rule

$$f(1) = 3$$

$$f(2) = 2$$

$$f(3) = 1$$

and let $g : \{1, 2, 3\} \rightarrow \{1, 2, 3\}$ be the function with rule

$$g(1) = 2$$

$$g(2) = 3$$

$$g(3) = 1.$$

Both f and g are permutations of $\{1, 2, 3\}$. Here's another way to write down the rules for these two permutations:

$$f = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \quad \text{and} \quad g = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}.$$

Each of these two functions is now specified by a matrix with the x -values in the top row and the (corresponding) y -values in the bottom row. So if, for example, we define another permutation

$$h = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 5 & 2 & 7 & 6 & 3 & 1 & 4 \end{pmatrix}$$

then we can see that $h(1) = 5$, $h(3) = 7$, and $h(6) = 1$.

Permutations play an important role in mathematics. In this course, we are especially interested in counting the number of permutations of a given finite set. Here is an example.

Example A.12 Determine the number of permutations of the set $\{1, 2, 3\}$.

Every permutation of $\{1, 2, 3\}$ can be written as a matrix with 1 2 3 at the top and the numbers 1, 2, and 3 in some order at the bottom. Since the set $\{1, 2, 3\}$ is small, with a little thought we can write all of its permutations down:

$$\begin{array}{lll} f_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} & f_2 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} & f_3 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \\ f_4 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix} & f_5 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} & f_6 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix} \end{array}$$

So there are six different permutations of the set $\{1, 2, 3\}$.

Suppose now that we want to work out the number of permutations of the set $\{a, b, c\}$. Although the elements of this set have different names to the elements in the set $\{1, 2, 3\}$, both these sets contain three elements. Clearly, the names of the elements of a set have no effect on the number of permutations of the set. Thus this set also has six permutations.

Example A.13 In Example A.12, we worked out the number of permutation of the set $\{1, 2, 3\}$ by writing all of them down. Here's another way: As we already mentioned, every permutation of $\{1, 2, 3\}$ is a matrix with 1 2 3 at the top and the numbers 1, 2, and 3 written in some order at the bottom. So the number of permutations of $\{1, 2, 3\}$ is just the number of different ways we can arrange the numbers 1, 2, and 3 at the bottom of the matrix, which we know from Theorem 39 is $3! = 6$.

The argument in the preceding example can be generalized in the following result:

Theorem 40. *The number of permutations of a finite set of n elements is $n!$.*

Example A.14 The number of permutations of the set

$\{\text{jacob zuma, julius malema, helen zille, mamphela ramphela, luis suarez}\}$

is $5! = 120$.

A.3 Falling factorials

From the previous two sections, we know that the number of ways of arranging n objects, or, equivalently, the number of permutations of a finite set of n objects, is $n!$. In this section we shall consider some similar counting problems. We begin with an illustrative example.

Example A.15 *Suppose that a group of six male MAM1000W students is writing a makeup test and that, before the test begins, four of them form a queue for the bathroom. How many different queues are possible?*

One of the six students must be first in the queue, one of the remaining five must then be second in the queue, one of other four must go third, and one of the last three must then go fourth (and last) in the queue. So the number of possible queues is

$$6 \cdot 5 \cdot 4 \cdot 3 = 360.$$

This looks sort of, but not quite, like what we saw in the section on factorials. With a little bit of algebraic trickery, we can rewrite the answer in terms of factorials:

$$6 \cdot 5 \cdot 4 \cdot 3 = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{2 \cdot 1} = \frac{6!}{2!}.$$

There's a $6!$ in this expression, which is nice because the total number of students is 6. The queue length is 4, though, so let's rewrite the answer so that the number 4 shows up as well:

$$6 \cdot 5 \cdot 4 \cdot 3 = \frac{6!}{2!} = \frac{6!}{(6-4)!}.$$

With the preceding example fresh in our minds, we make the following definition.

Definition A.16 If n and k are integers with $0 \leq k \leq n$, then we define the *falling factorial*

$$(n)_k = n(n-1) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$

Example A.17

$$(5)_2 = 5 \cdot 4 = \frac{5!}{(5-2)!} = 20.$$

Example A.18 Suppose that a group of n male MAM1000W students is writing a makeup test and that, before the test begins, k of them form a queue for the bathroom. How many different queues are possible?

One of the n students must be first in the queue, one of the remaining $n-1$ students must then be second, one of the other $n-2$ students must be third, and so on. When we choose the student to go last in the queue, there are already $k-1$ students in the queue, so there are $n-(k-1) = n-k+1$ students who can go last. Thus the total number of different queues is

$$n(n-1)(n-2) \cdots (n-k+1) = (n)_k = \frac{n!}{(n-k)!}.$$

The idea behind the preceding example can be stated more generally in the following way.

Theorem 41. The number of different ways of arranging k objects that have been chosen from a set of n objects is

$$(n)_k.$$

Notice that for the special case when $k = n$, Theorem 41 becomes Theorem 39, since

$$(n)_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!.$$

Example A.19 Suppose that ten MAM1000W students organize a drinking competition in which

the student who comes first gets a copy of all the MAM1000W tutorials, the student who finishes second gets a pen that was used to take notes during the lectures, and the student who finishes third gets to have lunch with Professor Gilmour. In how many different ways can the prizes be given out?

We are essentially asking how many different queues of 3 students we can form from the 10 (the student who is first in the queue comes first in the competition, the student who is second in the queue gets second prize, etc). So the number is

$$(10)_3 = \frac{10!}{(10-3)!} = 10 \cdot 9 \cdot 8 = 720.$$

Example A.20 Suppose that 9 male and 6 female MAM1000W students are writing a makeup test. In how many different ways can 3 of the male students and 2 of the female students form two queues for the bathrooms before the test begins?

The men can form $(9)_3$ different queues and the women $(6)_2$ different queues, so the total number of possibilities is

$$(9)_3 \cdot (6)_2 = \frac{9!}{6!} \cdot \frac{6!}{4!} = \frac{9!}{4!} = 1890.$$

Example A.21 How many different one-to-one functions are there from the set $\{a, b, c, d\}$ to the set $\{1, 2, 3, 4, 5, 6, 7\}$?

For each such function f , the quantities $f(a)$, $f(b)$, $f(c)$, and $f(d)$ must be numbers in the set $\{1, 2, 3, 4, 5, 6, 7\}$. $f(a)$ can be any of the numbers 1, 2, 3, 4, 5, 6, or 7. Since f is one-to-one, we can't have $f(b) = f(a)$, so whatever $f(a)$ is, $f(b)$ must be one of the remaining six numbers. Similarly, $f(c)$ can't be equal to $f(a)$ or $f(b)$, so whatever $f(a)$ and $f(b)$ are, it can be any of five different numbers. For the same reason, once the values of $f(a)$, $f(b)$, and $f(c)$ have been decided, $f(d)$ can be any of four numbers. It should be clear that this situation is one we've seen before: We're asking in how many different ways we can arrange four of the numbers from a set of seven. The answer is

$$(7)_4 = \frac{7!}{3!} = 840.$$

A.4 Combinations

When we're counting the number of ways we can arrange objects — like how many different queues of three students we can get from a group of seven — the order of the objects matters. The two queues

Aphiwe, Bandile, Caroline

and

Bandile, Aphiwe, Caroline

are clearly different. But what if we're counting something where the order of the objects doesn't matter? For example, if we write the names of the students in a MAM1000W tutorial group in a different order, it's still the same tutorial group. In the next example we discuss how to count things like tutorial groups, where the order of the objects doesn't matter.

Example A.22 *In how many different ways can a tutorial group of 3 students be chosen from a class of 5 students?*

For concreteness, let's give the students in the class names: Aphiwe, Bandile, Caroline, Dumisani, and Eve. If we were counting the number of queues of 3 students we could form from the students in this class, then the answer would be $(5)_3 = 5 \cdot 4 \cdot 3$. In a tutorial group, though, the order in which the students are listed doesn't matter:

Caroline, Bandile, Dumisani

and

Dumisani, Caroline, Bandile

are two different queues, but the same tutorial group. In fact, if we think about it, we can form $3! = 6$ different queues from these three students:

Queue 1: Bandile, Caroline, Dumisani
 Queue 2: Bandile, Dumisani, Caroline
 Queue 3: Caroline, Bandile, Dumisani
 Queue 4: Caroline, Dumisani, Bandile
 Queue 5: Dumisani, Bandile, Caroline
 Queue 6: Dumisani, Caroline, Bandile

There's nothing special about Bandile, Caroline, and Dumisani. If we have any tutorial group of 3 students, we can make $3!$ different queues from that tutorial group. Thus, the number of queues we can make from the whole class is $3!$ times the total number of tutorial groups (more precisely, the number of queues of length 3 we can make from Aphiwe, Bandile, Caroline, Dumisani, and Eve is $3!$ times the number of tutorial groups with three students). So the total number of tutorial groups is the total number of queues divided by $3!$. We already know that the total number of queues is $(5)_3$. Thus, the number of tutorial groups is

$$\frac{(5)_3}{3!} = \frac{5!}{(5-3)! 3!} = \frac{5!}{2! 3!} = 10.$$

Motivated by the previous example, we make the following definition.

Definition A.23 Let n, k be integers with $0 \leq k \leq n$. Then n choose k is the quantity

$$\binom{n}{k} = \frac{n!}{(n-k)! k!} = \frac{(n)_k}{k!}.$$

The quantity $\binom{n}{k}$ is also called a *binomial coefficient*. We shall see why later.

Example A.24

$$\binom{7}{4} = \frac{7!}{3! 4!} = 35.$$

In Example A.22, we found that the number of tutorial groups of 3 students we can make from a class of 5 is

$$\binom{5}{3} = \frac{5!}{2! 3!} = 10.$$

The same argument applies to any situation where it doesn't matter in which order we write the objects.

Theorem 42. *If the order of the objects doesn't matter, then the number of ways we can choose k objects from a set of n is $\binom{n}{k}$.*

Example A.25 *In how many different ways can we make a Monday Afternoon A tutorial group of 35 students from the 440 students in MAM1000W?*

This is a tutorial group, not a queue, so the order of the students doesn't matter. The answer is

$$\binom{440}{35} = \frac{440!}{405! 35!} = 8004707286270088181331734074011307208901475519813336.$$

You've probably seen the next definition before.

Definition A.26 If A and B are sets and every element of A is an element of B , then we say that A is a *subset* of B , written $A \subseteq B$.

Example A.27 If $A_1 = \{1, 3, 4\}$, $A_2 = \{2, 5\}$, and $B = \{1, 2, 3, 4\}$, then $A_1 \subseteq B$, but A_2 is not a subset of B (since $5 \in A_2$ but $5 \notin B$).

Example A.28 *How many subsets of 3 elements can we make from the set $\{a, b, c, d, e\}$?*

Remember that in a set, or subset, the order in which we write down the elements doesn't matter. For example, $\{a, c, e\}$ and $\{c, e, a\}$ are both the same subset of $\{a, b, c, d, e\}$. Since the order of the elements doesn't matter, the answer is

$$\binom{5}{3} = 10.$$

In the previous sections, we counted things, like queues and permutations, in which the order of the objects matters. In this section, we've been counting things like tutorial groups where the order of the objects doesn't matter. In general, a *combination* is a selection of a group of objects in which the order of the objects doesn't matter. Tutorial groups are combinations. Queues and permutations are not.

Example A.29 *Twelve male students are to write a MAM1000W makeup test. Before the test begins, four of them queue for the bathroom. While they're waiting for these students to come back, three of the remaining students decide to form their own study group. In how many different ways can all of this happen?*

The number of possible queues is $(12)_4$. A study group is like a tutorial group — it doesn't matter in which order we list the names of the students in it — so the number of possible study groups that the eight remaining students can form is $\binom{8}{3}$. The total number of different possibilities is therefore

$$(12)_4 \binom{8}{3} = \frac{12!}{8!} \frac{8!}{5! 3!} = 665280.$$

Remember that $\binom{n}{n}$ is the number of tutorial groups of size n we can pick from a class with n students, so it makes sense that $\binom{n}{n} = 1$. If you can convince yourself that there's only way we can pick a tutorial group containing *no* students from a class of n students, then it also makes sense that $\binom{n}{0} = 1$.

Another thing we notice from Pascal's Triangle is that each horizontal row is symmetric about its center. If we look at $n = 6$, for example, we see that

$$\begin{aligned}\binom{6}{0} &= 1 = \binom{6}{6} \\ \binom{6}{1} &= 6 = \binom{6}{5} \\ \binom{6}{2} &= 15 = \binom{6}{4}\end{aligned}$$

If we think about this some more, it seems that $\binom{6}{a} = \binom{6}{b}$ whenever $a + b = 6$ (like when $a = 2$ and $b = 4$). This is the same as saying that $\binom{6}{a} = \binom{6}{6-a}$. This makes sense: If we want to make a tutorial group of 2 students from a class of 6, then we can either list the 2 students in the tutorial group, or we can list the 4 students who are not in the tutorial group. So the number of ways we can choose 2 students from a class of 6 (to put into a tutorial group) is equal to the number of ways we can choose 4 students from a class of 6 (to not put into the tutorial group).

To prove that the binomial coefficients in general have this symmetry property requires a little proof.

Theorem 43. *If n and k are integers with $0 \leq k \leq n$, then*

$$\binom{n}{k} = \binom{n}{n-k}.$$

Proof.

$$\binom{n}{k} = \frac{n!}{(n-k)! k!} = \frac{n!}{k! (n-k)!} = \frac{n!}{(n-(n-k))! (n-k)!} = \binom{n}{n-k}.$$

□

Still another property of Pascal's Triangle is the following: Each number is the sum of the two numbers directly above it. For example, in the horizontal row $n = 6$ of Pascal's Triangle, the number 15 occurs twice. The first 'copy' of the number 15 is the binomial coefficient $\binom{6}{2}$.

The two numbers directly above it are 5 and 10, which are the binomial coefficients $\binom{5}{1}$ and $\binom{5}{2}$. So

$$\binom{6}{2} = \binom{5}{1} + \binom{5}{2}.$$

The relationship between these three binomial coefficients is an example of a more general property of all the binomial coefficients, which we now prove.

Theorem 44. *Let n and k be integers with $1 \leq k \leq n-1$. Then*

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Proof.

$$\begin{aligned} \binom{n-1}{k-1} + \binom{n-1}{k} &= \frac{(n-1)!}{(n-1-(k-1))! (k-1)!} + \frac{(n-1)!}{(n-1-k)! k!} \\ &= \frac{(n-1)!}{(n-k)! (k-1)!} + \frac{(n-1)!}{(n-k-1)! k!} \\ &= \frac{(n-1)!}{(n-k) \cdot (n-k-1)! (k-1)!} + \frac{(n-1)!}{(n-k-1)! k \cdot (k-1)!} \\ &= \frac{(n-1)!}{(n-k-1)! (k-1)!} \left[\frac{1}{n-k} + \frac{1}{k} \right] \\ &= \frac{(n-1)!}{(n-k-1)! (k-1)!} \cdot \frac{n}{k(n-k)} \\ &= \frac{n \cdot (n-1)!}{[(n-k) \cdot (n-k-1)!] [k \cdot (k-1)!]} \\ &= \frac{n!}{(n-k)! k!} \\ &= \binom{n}{k}. \end{aligned}$$

□

Example A.30 From the formula $\binom{n}{k} = \frac{n!}{(n-k)! k!}$, we have $\binom{8}{5} = 56$, $\binom{7}{5} = 21$, and $\binom{7}{4} = 35$. So, as we have just proved,

$$\binom{8}{5} = \binom{7}{4} + \binom{7}{5}.$$

The algebra in the proof of Theorem 44 is nasty. The proof, while correct, also provides no insight into why $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$. So here's a 'proof' using tutorial groups that illustrates *why* the theorem is true. This is an example of a 'Combinatorial Proof', in which we count the same thing in two different ways.

Example A.31 *In how many different ways can we make a Monday Afternoon A tutorial group of k students from the n students in MAM1000W?*

Let N be the answer (i.e., the number of different ways we can make a tutorial group of k students from a class of n students). We now work out the value of N in two different ways:

1. From Theorem 42, and many discussions, it should be clear that

$$N = \binom{n}{k}.$$

2. Let's suppose, without loss of generality, that Aphiwe is a member of the class. Then N is equal to the total number of tutorial groups (of k students) that contain Aphiwe plus the total number of tutorial groups (of k students) that do not contain Aphiwe.

- If a tutorial group does not contain Aphiwe, then all k of the students in that group must have been chosen from the $n - 1$ students in the class who are not Aphiwe. The number of tutorial groups that do not contain Aphiwe is therefore $\binom{n-1}{k}$.
- On the other hand, if a tutorial group does contain Aphiwe, then the remaining $k - 1$ students in that group must have been selected from the $n - 1$ students in the class who are not Aphiwe. So the number of tutorial groups that do contain Aphiwe is $\binom{n-1}{k-1}$.

Thus

$$N = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Finally, from our two calculations of the value of N , we have

$$\binom{n}{k} = N = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

A.6 The Binomial Theorem

Let's expand $(x + y)^n$ for a few small values of n :

$$\begin{aligned} n = 0 : & \quad (x + y)^0 = 1 \\ n = 1 : & \quad (x + y)^1 = x + y \\ n = 2 : & \quad (x + y)^2 = (x + y)(x + y) = y^2 + 2xy + x^2 \\ n = 3 : & \quad (x + y)^3 = (x + y)(x + y)^2 = y^3 + 3xy^2 + 3x^2y + x^3 \\ n = 4 : & \quad (x + y)^4 = (x + y)(x + y)^3 = y^4 + 4xy^3 + 6x^2y^2 + 4x^3y + x^4 \end{aligned}$$

The coefficients of the polynomial on the right hand side of the $n = 4$ row are

$$1 \quad 4 \quad 6 \quad 4 \quad 1$$

This is the same as the $n = 4$ row in Pascal's Triangle (page 71). Since the numbers in the $n = 4$ row of Pascal's Triangle are the binomial coefficients $\binom{4}{k}$, this means that

$$(x + y)^4 = \binom{4}{0}y^4 + \binom{4}{1}xy^3 + \binom{4}{2}x^2y^2 + \binom{4}{3}x^3y + \binom{4}{4}x^4.$$

If you compare the other rows of Pascal's Triangle with $(x + y)^n$, you'll see the same thing:

$$\begin{aligned} (x + y)^0 &= \binom{0}{0} \\ (x + y)^1 &= \binom{1}{0}y + \binom{1}{1}x \\ (x + y)^2 &= \binom{2}{0}y^2 + \binom{2}{1}xy + \binom{2}{2}x^2 \\ (x + y)^3 &= \binom{3}{0}y^3 + \binom{3}{1}xy^2 + \binom{3}{2}x^2y + \binom{3}{3}x^3 \end{aligned}$$

That this is true for every nonnegative integer n is the *Binomial Theorem*, which we prove next. It's also why the quantity $\binom{n}{k}$ is called a *binomial coefficient* — the expression $x + y$ is called a *binomial*, so the binomial coefficients are the numbers that show up when we expand a power of a binomial.

Theorem 45 (Binomial Theorem). *For all real numbers x, y and every nonnegative integer n ,*

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Proof. Our proof will be by induction on n . When $n = 0$, the left hand side is

$$(x + y)^0 = 1$$

while the right hand side is

$$\sum_{k=0}^0 \binom{0}{k} x^k y^{-k} = \binom{0}{0} x^0 y^0 = 1.$$

This proves the Base Case. Suppose now that the statement is true for some nonnegative integer n , i.e., that

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}.$$

Then

$$\begin{aligned} (x + y)^{n+1} &= (x + y)(x + y)^n \\ &= (x + y) \sum_{k=0}^n \binom{n}{k} x^k y^{n-k} \quad (\text{by the Inductive Hypothesis}) \\ &= \sum_{k=0}^n \binom{n}{k} x^{k+1} y^{n-k} + \sum_{k=0}^n \binom{n}{k} x^k y^{n-k+1} \\ &= \sum_{k=0}^n \binom{n}{k} x^{k+1} y^{(n+1)-(k+1)} + \sum_{k=0}^n \binom{n}{k} x^k y^{n+1-k} \end{aligned}$$

To make any further progress, we have to get the terms in the two sums on the right hand side to have the same powers of x and y . We do this by a change of summation index in the first sum: Let $t = k + 1$. Then when $k = 0$, $t = 1$, and when $k = n$, $t = n + 1$. Thus,

$$(x + y)^{n+1} = \sum_{t=1}^{n+1} \binom{n}{t-1} x^t y^{n+1-t} + \sum_{k=0}^n \binom{n}{k} x^k y^{n+1-k}$$

Now we want to bring the two sums on the right hand side back together. To do so, remember that a summation index is a dummy variable — its name has no effect on the sum². Thus we can replace the summation index t in the first sum with k , so that the two sums use the same summation index:

$$\begin{aligned} (x + y)^{n+1} &= \sum_{k=1}^{n+1} \binom{n}{k-1} x^k y^{n+1-k} + \sum_{k=0}^n \binom{n}{k} x^k y^{n+1-k} \\ &= \sum_{k=1}^n \binom{n}{k-1} x^k y^{n+1-k} + \binom{n}{(n+1)-1} x^{n+1} y^{n+1-(n+1)} + \binom{n}{0} x^0 y^{n+1-0} + \sum_{k=1}^n \binom{n}{k} x^k y^{n+1-k} \\ &= \binom{n+1}{0} x^0 y^{n+1} + \sum_{k=1}^n \left[\binom{n}{k-1} + \binom{n}{k} \right] x^k y^{n+1-k} + \binom{n+1}{n+1} x^{n+1} y^0 \end{aligned}$$

²For example, $\sum_{i=1}^3 i^2 = 14 = \sum_{j=1}^3 j^2 = \sum_{k=1}^3 k^2$.

To get the last line, we used the fact that $\binom{n+1}{n+1} = 1 = \binom{n}{n}$ and $\binom{n+1}{0} = 1 = \binom{n}{0}$. Proceeding, we know from Theorem 44 that when $1 \leq k \leq n$, we have $\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$. Thus

$$\begin{aligned}(x+y)^{n+1} &= \binom{n+1}{0}x^0y^{n+1} + \sum_{k=1}^n \binom{n+1}{k}x^ky^{n+1-k} + \binom{n+1}{n+1}x^{n+1}y^0 \\ &= \sum_{k=0}^{n+1} \binom{n+1}{k}x^ky^{n+1-k}.\end{aligned}$$

We have proved that the statement is true when $n = 0$. We have proved that whenever the statement is true for some integer $n \geq 0$, it is also true for the integer $n + 1$. Thus, by the Principle of Mathematical Induction, the statement is true for every nonnegative integer n . $\square$

Example A.32

$$\begin{aligned}(x+y)^3 &= \sum_{k=0}^3 \binom{3}{k}x^ky^{3-k} \\ &= \binom{3}{0}y^3 + \binom{3}{1}xy^2 + \binom{3}{2}x^2y + \binom{3}{3}x^3 \\ &= y^3 + 3xy^2 + 3x^2y + x^3\end{aligned}$$

Example A.33 Suppose we want to expand $\left(2a - \frac{b}{a}\right)^5$. Using the Binomial Theorem with $x = 2a$ and $y = -\frac{b}{a}$, we find that

$$\begin{aligned}\left(2a - \frac{b}{a}\right)^5 &= \sum_{k=0}^5 \binom{5}{k}(2a)^k \left(-\frac{b}{a}\right)^{5-k} \\ &= \sum_{k=0}^5 \binom{5}{k}2^ka^k(-1)^{5-k}b^{5-k}a^{k-5} \\ &= \sum_{k=0}^5 \binom{5}{k}(-1)^{5-k}2^ka^{2k-5}b^{5-k} \\ &= -\binom{5}{0}a^{-5}b^5 + \binom{5}{1}2a^{-3}b^4 - \binom{5}{2}4a^{-1}b^3 + \binom{5}{3}8ab^2 - \binom{5}{4}16a^3b + \binom{5}{5}32a^5 \\ &= -a^{-5}b^5 + 10a^{-3}b^4 - 40a^{-1}b^3 + 80ab^2 - 80a^3b + 32a^5\end{aligned}$$

Appendix B

Some pictures of applications of graph theory





Figure B.1: A graph with ten million edges. Paul Butler was an intern at Facebook when he created this visualisation of Facebook friendships [But10]. *‘I began by taking a sample of about ten million pairs of friends from Apache Hive, our data warehouse. I combined that data with each user’s current city and summed the number of friends between each pair of cities. Then I merged the data with the longitude and latitude of each city.’*

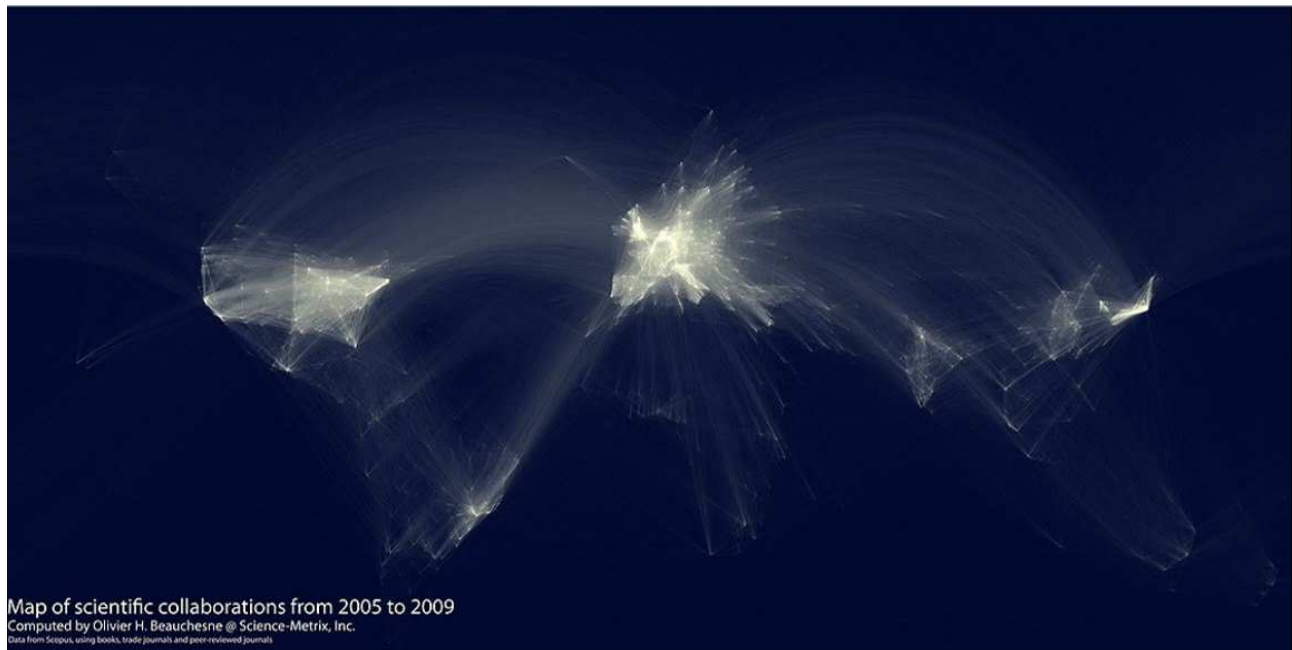


Figure B.2: Inspired by Paul Butler’s Facebook picture, Olivier Beauchesne [Bea11] used a database from a company called Science-Metrix to visualise scientific collaboration (above). *‘From this database, I extracted and aggregated scientific collaboration between cities all over the world. For example, if a UCLA researcher published a paper with a colleague at the University of Tokyo, this would create an instance of collaboration between Los Angeles and Tokyo. The result of this process is a very long list of city pairs, like Los Angeles-Tokyo, and the number of instances of scientific collaboration between them. Following that, I used the geoname.org database to convert the cities’ names to geographical coordinates.’* Below, a close-up of South Africa.

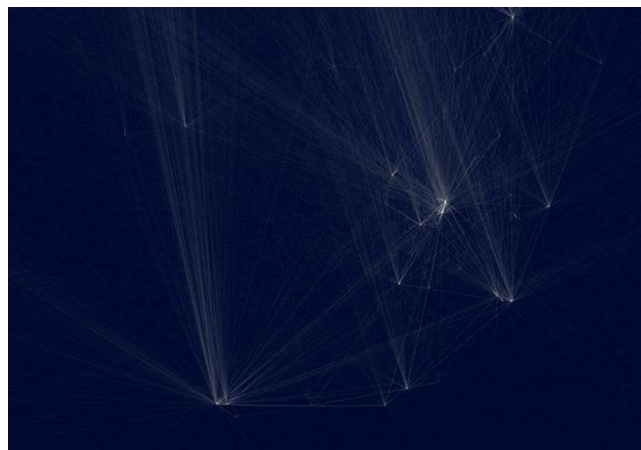


Figure B.3 shows a graph labeled ∂ on the left, followed by an equals sign and a large bracketed expression of nine graphs with various plus and minus signs, ending with an equals zero. Each graph is a circular diagram with 13 internal nodes and 8 external nodes labeled 1-8. Some nodes are shaded blue, and some edges are red. Below each graph is a set of numbers in curly braces, with some numbers in red.

The graphs and their associated sets of numbers are:

- Graph 1 (top left): $\{4,7,6,9,8,11,10,13\}$
- Graph 2 (top middle): $\{4,7,6,9,10,8,11,13\}$
- Graph 3 (top right): $\{4,8,6,9,7,10,11,13\}$
- Graph 4 (middle left): $\{7,4,6,9,8,10,11,13\}$
- Graph 5 (middle middle): $\{3,7,6,9,8,10,12,13\}$
- Graph 6 (middle right): $\{4,7,6,9,8,10,13,11\}$
- Graph 7 (bottom left): $\{4,7,6,10,8,9,11,13\}$
- Graph 8 (bottom middle): $\{4,7,5,9,8,10,11,14\}$
- Graph 9 (bottom right): $\{4,7,9,6,8,10,11,13\}$

Figure B.3: The graphs here are ‘on-shell diagrams’ used for calculations in theoretical high energy physics [AHBC⁺].

Figure B.4: The Algerian electrical network, pre-1997. ‘*This paper presents solution of optimal power flow (OPF) problem of a power system via an Ant Colony Optimization Meta-heuristic method. The objective is to minimize the total fuel cost of thermal generating units and also conserve an acceptable system performance in terms of limits on generator real and reactive power outputs, bus voltages, shunt capacitors/reactors, transformers tap-setting and power flow of transmission lines. Simulation results on the Algerian Electrical Network show that the Ant Colony Optimization method converges quickly to the global optimum.*’ [BS05]

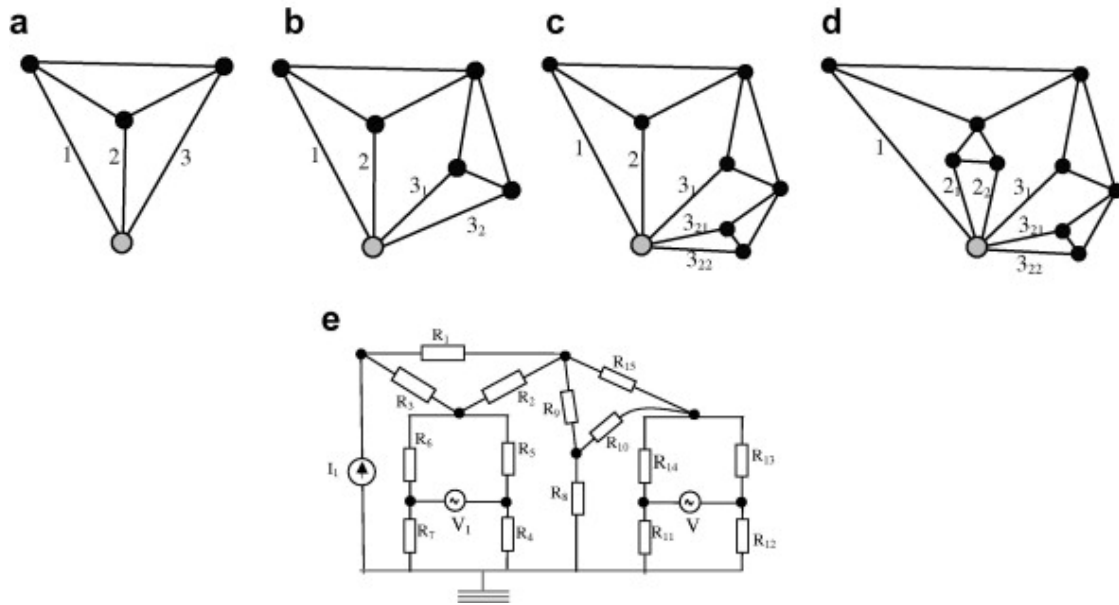


Figure B.5: Using graphs to design an electrical circuit. ‘Since networks, whose corresponding graphs are self-dual, possess the interchangeability property of currents and voltages, in accordance to the result devised in Section 4.1, the procedure of expansion can be employed for constructing multi-functional electrical circuits’ [STR08]

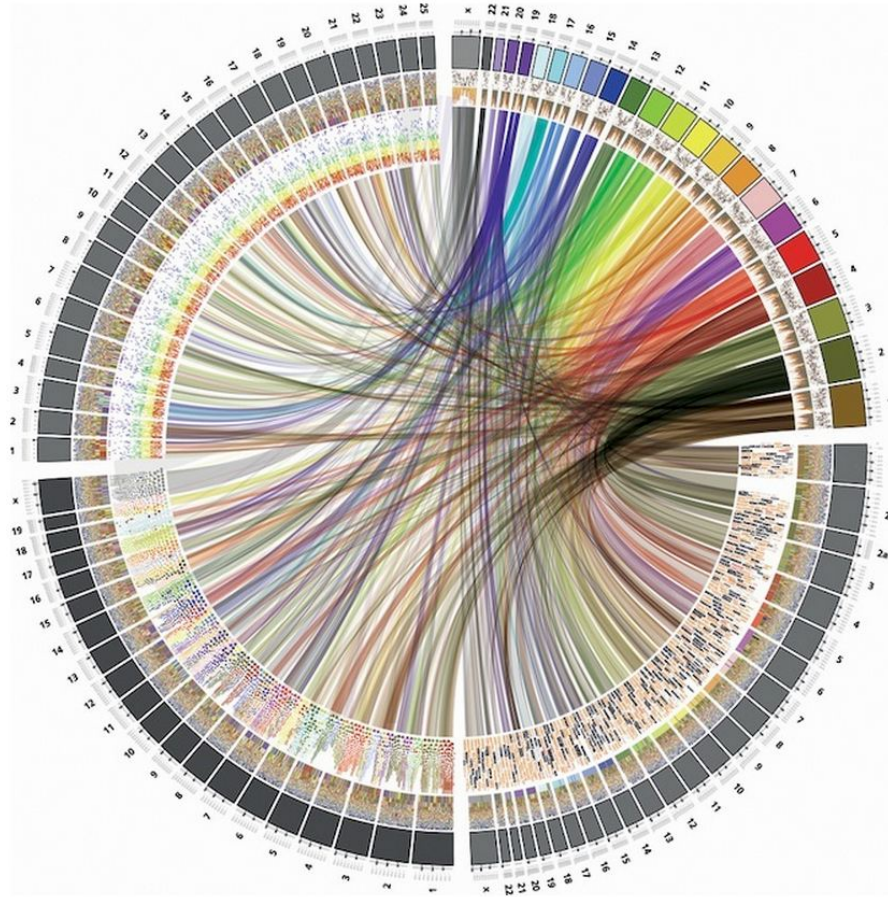


Figure B.6: Visualising biological data: ‘Clockwise from top right, the genomes of a human, a chimpanzee, a mouse and a zebrafish are arranged in a circle, with each color square corresponding to a pair of chromosomes. Lines connect similar DNA sequences, visually emphasizing just how much DNA we share with other species.’ The image was produced by Martin Krzywinski (<http://mkweb.bcgsc.ca/>) for EMBO. The explanation is from an article [Kei13] in the magazine Wired.

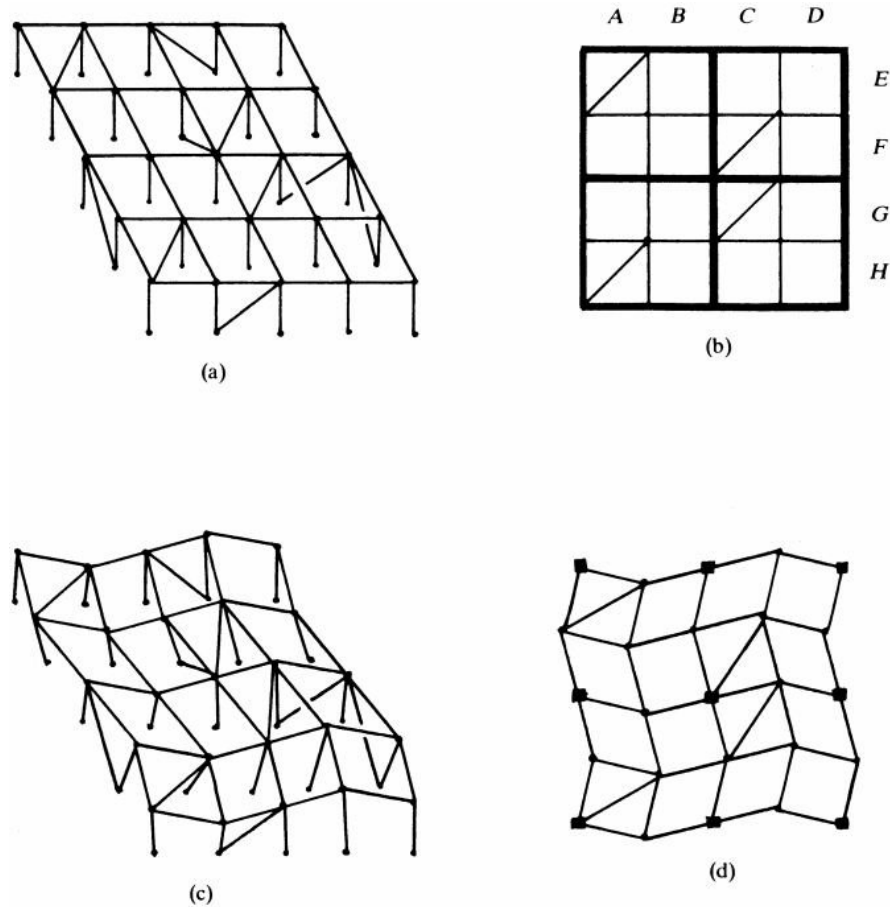


Figure B.7: Bracing rectangular frameworks. *‘This paper describes the economical placing of diagonal braces in the walls and ceiling of a rectangular one story building. It begins with the definition of the structure geometry of a graph embedded in Euclidean space: a combinatorial geometry (matroid) on the set of potential braces. When the embedded graph is a plane grid of squares the geometry is graphic. Then, for example, minimal rigidifying sets of braces correspond to spanning trees in a complete bipartite graph. The methods used in the plane case are extended to analyze how sets of wall and ceiling braces in a one story building can be dependent’.* [BC79]

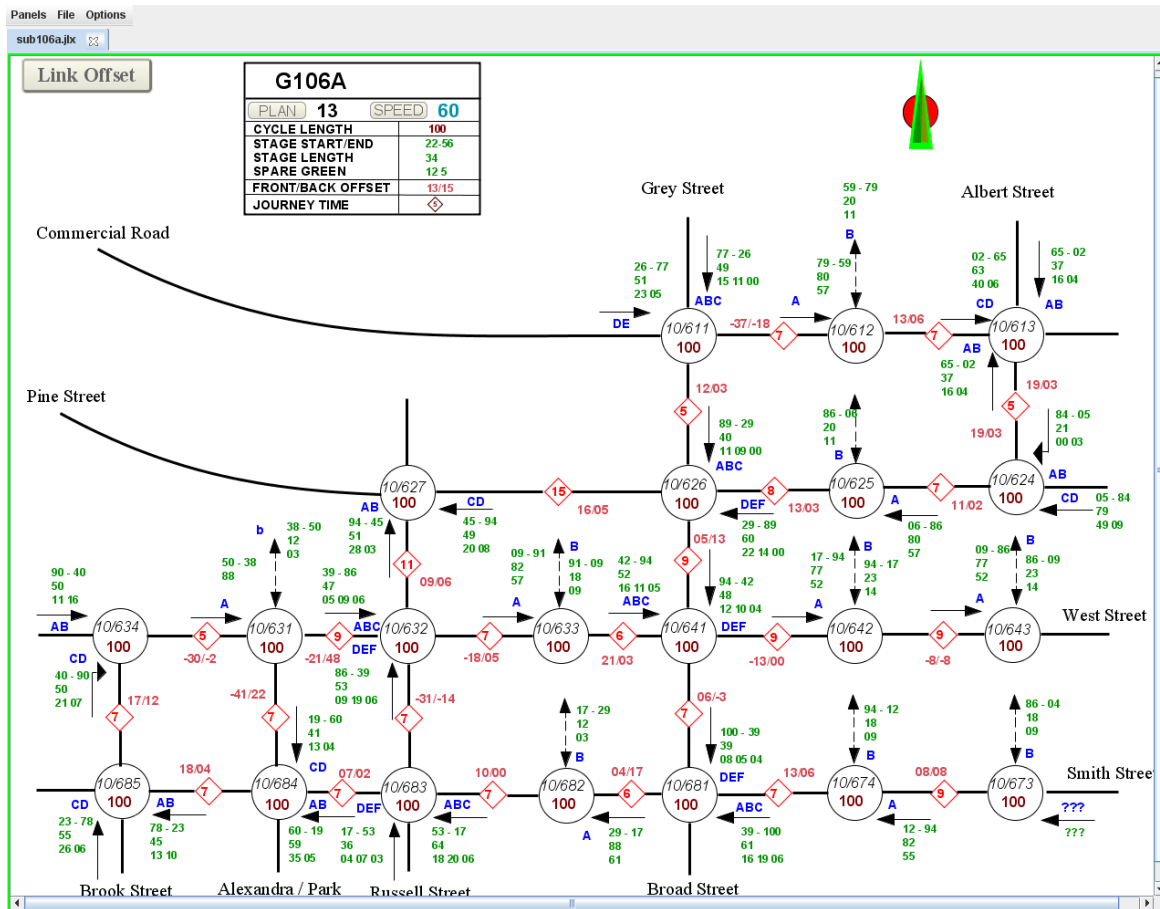


Figure B.8: Traffic light phasing in Durban (image courtesy of eThekweni Transport Authority). How should green times be assigned to a group of traffic lights?

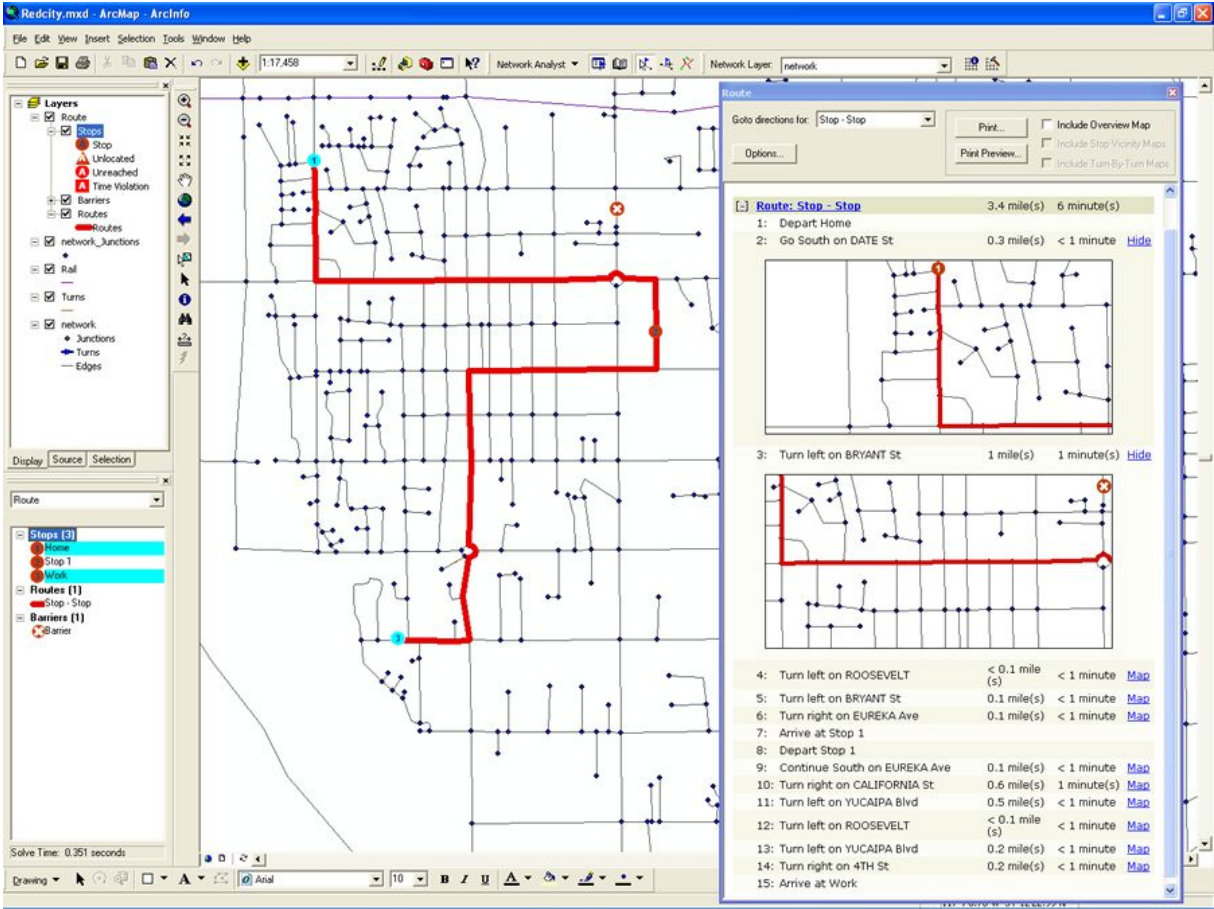


Figure B.9: Driving directions (from www.esri.com)

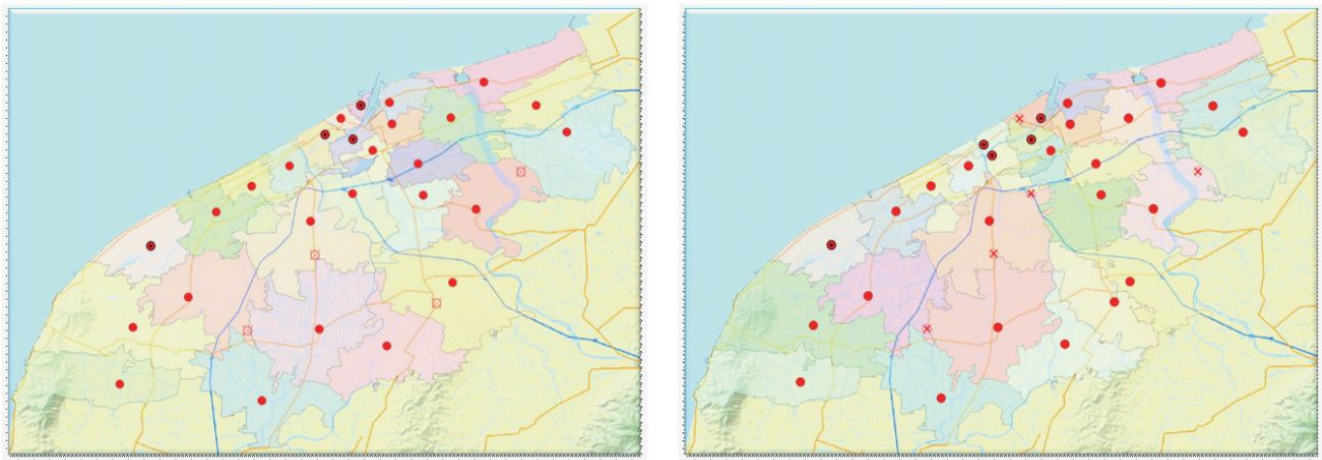


Figure 3. Current and future optimal allocation of 27 ambulances with their catchment areas.
 Note: (a) current allocation, (b) future allocation in 2030, (•) indicates optimal locations that are the same as current locations, (⊗) current sites not selected by the optimisation and (•) new locations.

Figure B.10: Facility location problems like the above [SCSB09] have been the subject of a lot of research in graph theory.

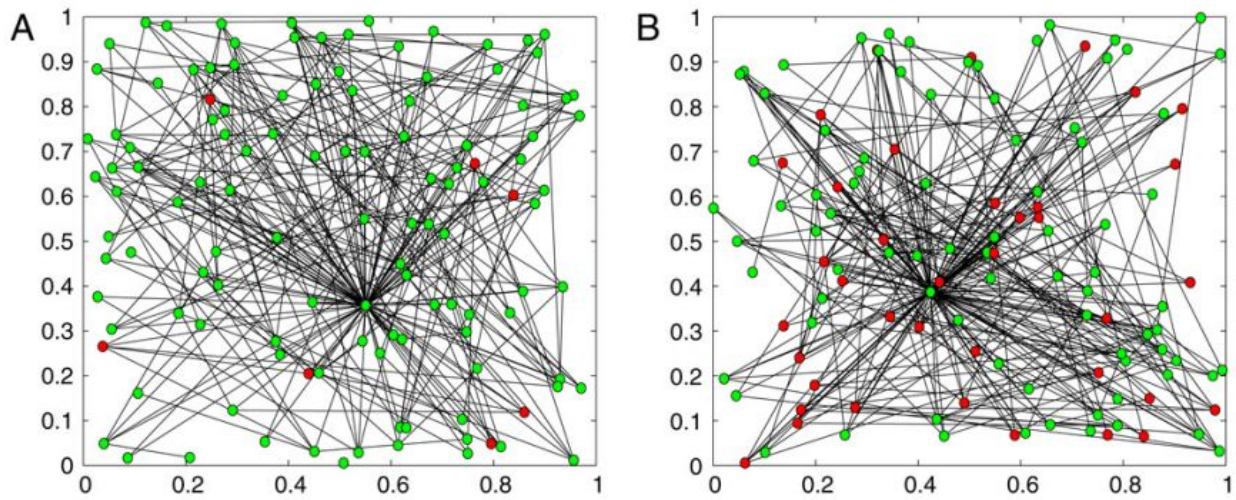


Fig. 4. Shows a situation where a single infective (A: reasonable and B: unreasonable) is introduced into the population. The parameter values are given by $\rho = 0.2$, $\gamma = 0.8$, $\beta = 0.5$ for A and $\beta = 0.9$ for B.

Figure B.11: Modelling the spread of tuberculosis [NMR09].

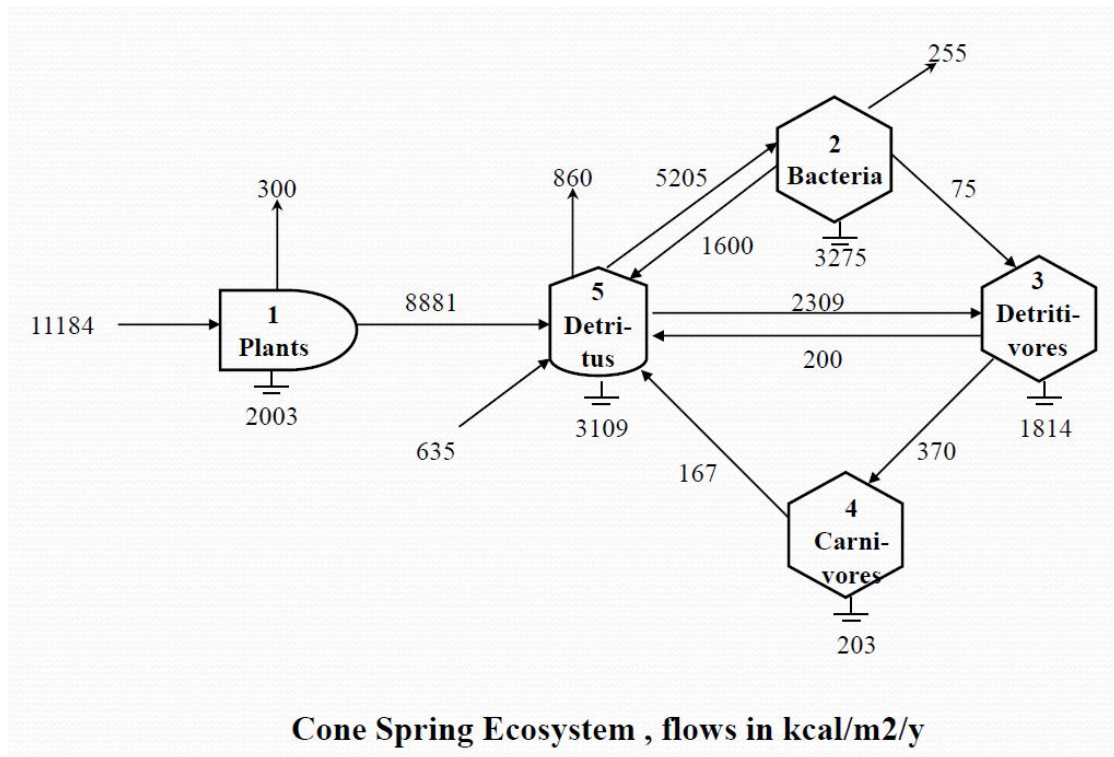


Figure B.12: When is an ecosystem stable? Ecologists model ‘foodwebs’, like the above [Sch11], using weighted directed graphs.

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